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The Pedestrian Gap: How Sidewalk Policy Obstructs Mobility Over Time

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Executive Summary

Project Title: *The Pedestrian Gap: How Sidewalk Policy Obstructs Mobility Over Time*

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Deliverable: [*Mind the Gap: Lessons from Denver to Rebuild Your Sidewalk Network*](#)

Problem. Forty-three percent of Denver's sidewalk network is missing or substandard. Most American cities are in a similar position because most use the same legal model: sidewalks sit in the public right-of-way, but the duty to build and maintain them rests with adjacent property owners. The model is cheap to administer and easy to defend in court, but it produces fragmented networks, slow repair cycles, and growing legal exposure. The gap costs more to close each year it goes unaddressed.

Context. Denver's 2022 voter approval of Initiative 307 replaced the property-owner obligation with a dedicated parcel fee funding a nine-year, citywide repair and construction program. The reform took more than twenty years to reach the ballot and makes Denver one of the few large U.S. cities to have completed the transition. Developed in partnership with Fehr & Peers, this capstone documents Denver's path and compares it to Seattle and Minneapolis to extract lessons for cities considering the same change.

Findings. The analysis combined municipal sidewalk inventories with development-era classifications, HOLC redlining maps, historic streetcar alignments, and municipal policies. Three findings frame the deliverable:

- *Path dependency drives the gap.* The car era (1940–1990) is the single largest era category for missing sidewalks in all three cities: 35 percent in Denver, 41 percent in Minneapolis, and 74 percent in Seattle. Pre-1940 areas account for only 15 to 18 percent in each city; the remaining gaps fall on parcels too recent, too non-residential, or too undocumented to classify by era. Car era subdivisions came in under permissive standards that treated sidewalks as optional, and retrofitting them today costs several times more than original installation because curb-and-gutter, drainage, and right-of-way conditions must all be reworked.
- *The equity story is complicated.* In Denver and Minneapolis, formerly redlined HOLC C and D areas have disproportionately high sidewalk gap rates. But the largest share of missing sidewalk in Denver (61 percent) and Seattle (62 percent) falls in HOLC-ungraded areas, including land that was undeveloped in the 1930s and later built out as car-era suburbs. Prioritization frameworks that focus only on

redlined neighborhoods may miss the larger share of the pedestrian network deficit.

- *Minor reform within the property-owner model cannot be relied on to close the gap.* Denver's 2017–2020 Neighborhood Sidewalk Repair Program inspected on a rotating cycle, made repairs, and assessed adjacent owners. A 2020 City Audit projected the program would take more than 400 years to finish at that pace. Minneapolis operates a structurally similar inspection-and-bill program today and has not closed many gaps in recent years. Simply improving how a city administers the property-owner model is not a solution to the problems municipal sidewalk networks face.

Recommendations. Our deliverable, *Mind the Gap*, presents twelve municipal programs in two tiers: programs that retain property-owner liability and programs that transfer cost, liability, or both onto a public or collective mechanism. Three takeaways follow:

1. *Sidewalk governance needs to be designed, not accepted.* The private-liability model evolved from nineteenth-century police-power doctrine until its current form was adopted by most US cities throughout the mid twentieth-century. Denver's 1894 Charter describes functional Sidewalk Districts with width standards exceeding current ADA requirements. However, that system was dismantled following explosive post-war growth.
2. *Distinguish operational reform from structural reform.* Inspection-and-bill, shared-cost, and point-of-sale programs improve delivery but keep property owners on the hook. Utility fees, dedicated parcel fees, millages, parcel taxes, sidewalk improvement districts, and general-fund programs replace the model. Both are available; only the second changes the outcome.
3. *Design equity as a multi-layer lens.* Historic redlining, the car era gap in ungraded areas, and present-day access to transit, schools, and essential services each identify a different set of priority neighborhoods. The funding mechanism's distributional design (flat fee, tiered fee, rebate) is a separate decision layered on top that determines if a program is equitable or not.

Final thoughts. Denver's reform arc is the most thorough contemporary case, but it is not the only path. Berkeley and Ann Arbor reached structural reform through voter-approved tax measures; Payette, Englewood, and Cheney did so through council action; Seattle delivers interim pedestrian access through its walkway program while maintenance liability remains private. *Mind the Gap* is built to help cities locate themselves on that spectrum and identify a next step.

The guide is hosted at sidewalkplanner.github.io/mind-the-gap.

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Chapter 1: Introduction

1.1 Background

Denver's sidewalk system reflects more than a century of shifting development patterns and public policy decisions. The City has long operated under a fragmented framework in which sidewalk construction, maintenance, and replacement were largely the responsibility of adjacent property owners. This approach produced uneven outcomes across neighborhoods and limited the City's ability to treat sidewalks as a cohesive transportation network. In 2022, voters approved Initiative 307, also known as *Denver Deserves Sidewalks*, establishing a new funding mechanism and shifting the City towards a publicly managed sidewalk network.

This recent transition represents a structural shift in the responsibility for Denver's sidewalks from a decentralized, property-based obligation towards a coordinated municipal network. Denver's experience provides a contemporary context for examining how governance structures, development history, and funding models shape sidewalk networks over time.

1.2 Problem Statement

The City and County of Denver faces a significant challenge: forty-three percent of the city's sidewalk network is either missing or not wide enough to meet modern accessibility standards. For decades, and like many cities, Denver has operated under a "private liability" model, treating sidewalks as a private asset rather than a public utility. This policy's legacy has resulted in a disjointed network, with pockets of inaccessibility where the ability to safely traverse a region on foot is determined by the historical context in which a neighborhood was built out and relied on maintenance of individual property owners. *Figure 1* maps the density of sidewalk gaps across Denver, showing where missing sidewalk segments are most concentrated as of February 2025.

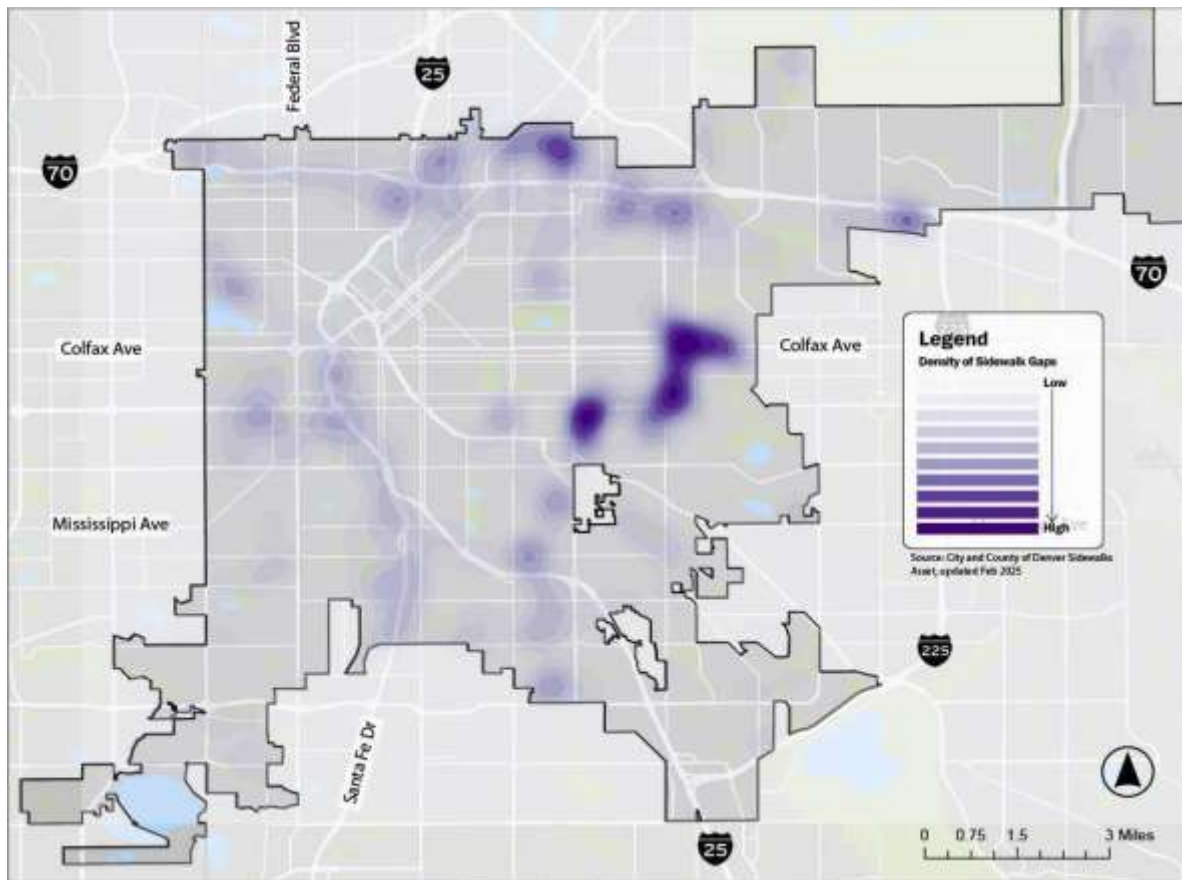


Figure 1. Sidewalk Gap Density: Denver

As identified through spatial analysis of Denver and peer cities, the shift from streetcar suburb typologies to autocentric development in the mid-20th century explicitly deprioritized pedestrian infrastructure's contiguousness and usefulness in favor of wider roads and non-walkable communities. This transition has created gaps in the sidewalk system; today, the cost of retrofitting sidewalks is prohibitively high due to larger parcel setbacks, differences in material conditions, and individual parcel landscaping constraints. Subject matter experts emphasize the logistical failure inherent in this system as well as the systemic barriers; the current state of the sidewalk network reflects a prioritization of liability avoidance over the human experience and public safety.

While the *Denver Deserves Sidewalks* reform marks a shift toward municipal management of sidewalks, the physical form of the city and the legacy of prior policy decisions continue to shape what is feasible today. Sidewalk conditions, materials, and network continuity vary significantly by development era, reflecting different regulatory environments and infrastructure standards. Without accounting for these historical patterns, planners may adopt uniform implementation strategies that overlook the structural constraints embedded in different neighborhood typologies.

This project responds to that challenge by synthesizing historical policy analysis and spatial data into a policy guide for cities aspiring to take back responsibility for the

pedestrian realm. Using Denver as a primary case study, the guide is designed to help municipalities evaluate governance structures, funding models, and development-era conditions as they consider transitions toward coordinated, publicly managed pedestrian systems.

1.3 Client Overview

The client for this capstone project is Fehr & Peers, a leading transportation planning and engineering firm with a presence in the Rocky Mountain region and across the United States. Established in 1985, the firm specializes in providing data-driven solutions to transportation challenges, ranging from multimodal corridor studies to long-range regional plans. In the context of Denver, Fehr & Peers has been a vital partner to the City and County of Denver in the development of the Sidewalk Implementation Plan. Their expertise in network analysis has provided the technical foundation for the city's current effort to address its sidewalk gaps.

1.4 Project Purpose & Objectives

The purpose of this capstone project is to produce a sidewalk network reform guide that helps municipalities evaluate and restructure their approach to pedestrian infrastructure. Developed in partnership with Fehr & Peers, this guide complements the firm's technical work on Denver's Sidewalk Implementation Plan by extending the lessons of Denver's governance transition into a replicable framework for other cities. The specific objectives of this project are:

- To produce a Sidewalk Network Reform Guide that provides municipalities with an actionable framework for transitioning from private-liability sidewalk models to publicly managed pedestrian networks.
- To draw from case studies of cities at different stages of sidewalk network reform transferable policy lessons for governance structures, funding models, and implementation strategies.
- To determine whether the era in which communities were developed serves as a reliable predictor of sidewalk network fragmentation across peer cities, and to identify how era-specific conditions shape the cost and feasibility of infrastructure retrofits.

This guide is intended for municipal planners, transportation engineers, advocacy organizations, and elected officials in cities still operating under private-liability sidewalk models and considering a transition toward publicly coordinated management. The guide does not provide a scoring methodology for project prioritization, prescribe specific construction standards, or generate cost estimates; those decisions are necessarily city-specific and depend on local conditions, budgets, and political context that no single guide can anticipate. Rather, the guide focuses on the upstream governance decisions that must be resolved before implementation planning can begin.

Chapter 2: Background Research & Literature Review

2.1 Introduction to Literature

The literature surrounding municipal infrastructure highlights clear tensions between historic urban form and modern mobility requirements. To understand gaps in pedestrian infrastructure, this research synthesizes three primary bodies of work: the evolution of sidewalk governance, an analysis of the mode shift from transit to autocentric mobility, and the legal/financial legacies of discriminatory housing policies. By examining these themes, the literature review provides the framework needed to evaluate how the existing sidewalk challenges were constructed and maintained.

This first review explores the shift in sidewalk responsibility, tracing the transition from early municipal management to the private liability models that are common in US cities today. We then researched the work of urban historians to categorize the built environment into distinct development eras: the pre-1940s “pre-car era” and the post-1940s - 1990s “car era” serving as a primary analysis framework. Finally, the literature incorporates modern analysis of systemic disinvestment to explain how retrofitting pedestrian infrastructure remains a significant challenge.

2.2 Theme 1: Evolution of Sidewalk Governance

Sidewalks are in a unique legal position in American municipal governance. They are located within the public right-of-way, yet responsibility for sidewalk construction and regular maintenance historically falls on abutting property owners. This structure, often described as “private maintenance within public space,” emerged from nineteenth-century interpretations of municipal police power and special assessment authority (Frug & Barron, 2013).

Under the police power doctrine, municipalities are authorized to regulate factors like public health, safety, and welfare. Courts have consistently upheld local ordinances requiring abutting property owners to construct and maintain sidewalks as a legitimate exercise of that power (*Gonzales v. San Jose*, 2004). This is often justified because municipalities deem sidewalks as a “special benefit” to adjacent parcels by improving access, drainage, and property value. Because of these benefits, courts have allowed municipalities to shift maintenance costs to individual owners through various means.

However, maintenance duty and tort liability have been treated differently. Even where ordinances required owners to repair sidewalks, municipalities remained liable for injuries occurring within the right-of-way. *Gonzales* clarifies this distinction: while cities can mandate property-owner maintenance obligations, they cannot use those ordinances to transfer tort liability to owners. Courts reasoned that because sidewalks remained public infrastructure, ultimate responsibility for safe passage rested with the city. This meant that property owners bore maintenance obligations, but cities faced liability risk if defects persisted. In practice, this encourages reactive enforcement systems. Rather than maintaining sidewalks through proactive management, many cities have opted to instead inspect sidewalks and cite owners for safety or maintenance infractions.

The Americans with Disabilities Act (ADA) of 1990 added another layer of legal complexity to sidewalk regulation. The ADA, particularly Title II, imposed affirmative obligations on public entities to ensure accessible pedestrian infrastructure. Federal courts have repeatedly held that sidewalks constitute a “service, program, or activity” of local government and must meet the standards of the ADA, such as in *Barden v. City of Sacramento* (2002). This solidified a shift from discretionary maintenance toward enforceable legal compliance. Municipalities could no longer rely solely on abutting owners for repairs and maintenance. If the City is made aware of deficiencies, they must address them.

The consequences of this deferred maintenance model are most starkly illustrated by Los Angeles, where decades of neglect led to the landmark class action *Willits v. City of Los Angeles* (2016), in which disability rights advocates alleged that the city’s crumbling sidewalks violated the ADA by denying access to approximately 280,000 residents with mobility disabilities. The resulting settlement – requiring \$1.37 billion in repairs over 30 years – stands as the largest disability access class action settlement in U.S. history and underscores how prolonged reliance on property-owner maintenance can expose municipalities to significant legal and financial liability.

The division of responsibility between municipalities and abutting property owners has been further shaped by state-level municipal governance. Generally, municipalities only have the powers explicitly granted to them by the state. This means that sidewalk regulation and assessment authority require clear statutory authorization. In contrast, municipalities in home rule states, such as Colorado, have broader authority to structure maintenance and funding models as they see fit.

2.3 Theme 2: The Geometry of Inaccessibility

The physical pattern of Denver’s sidewalk network is the legacy of various development eras, each with its own assumptions about how people would move through the city. The following sections trace this evolution and examine why the gaps left behind are so difficult to retrofit today.

2.3.1 The Streetcar Suburb (Pre-1940s)

In his seminal book, *Streetcar Suburbs* (1978), Sam Bass Warner contrasts the streetcar era developments of the nineteenth century with the auto-centric suburbs that followed. Warner argues that the pre-automobile city was organized around mixed, mutually reinforcing land uses, with the built environment shaped in response to the daily needs of its residents.

Neighborhoods of this era were built at a pedestrian scale. Commercial retail was concentrated around streetcar stops, both multifamily and single-family housing filled in alongside the lines, and an almost entirely contiguous sidewalk network tied these uses together. In Warner’s framing, sidewalks were an essential transportation network that allowed dense populations to reach the transit system and amenities it needed. This is in stark contrast to auto-centric developments which often focused on a single mode of travel.

2.3.2 The Automobile Suburb (Post-1940s)

The fragmentation of the pedestrian network is a direct result of the transition from what Martin Melosi (2004) defines as the "walking city" to the "automobile city." In the earlier era, urban form was dense, and contiguous sidewalks were needed for daily life and transit access. As the automobile became the dominant transportation mode, city design changed. Melosi (2004) describes the walking city as compact, with intermingled residences and workplaces, and the automobile city as a postwar form that dispersed population into the suburbs and eroded the patterns of the streetcar city. In older neighborhoods, contiguous sidewalks are common, reflecting a pre-car era organized around pedestrian-scale movement.

In areas primarily developed in the car era, there are noticeable gaps in the sidewalk network where sidewalks became a limited amenity rather than an expected community resource. Development patterns shifted to spread-out single-family homes, narrow sidewalks, and businesses concentrated on arterial roads, because walking was no longer the primary travel mode. As walking became a leisure activity rather than a mode of transportation, the infrastructure that supported it was systemically left out of development plans.

2.3.3 The Retrofit Challenge

In car-centric developments, whether sidewalks were never developed or were limited to narrow curb-attached strips, these design choices were frequently implemented to reduce initial capital costs. This created what urban theorists call 'path dependency' (Dunham-Jones and Williamson, 2011), meaning those previous decisions on how sidewalk networks were or weren't laid have implications for modern users. As Dunham-Jones and Williamson (2011) argue, once a suburban typology is established without pedestrian infrastructure, the cost and complexity of retrofitting that environment increases over time. In car era developments, retrofitting sidewalks is rarely a matter of pouring concrete. These areas lack the necessary curb-and-gutter systems, established drainage patterns, and public right-of-way width found in earlier grid-based developments. A single block of retrofit can require massive stormwater management overhauls that were not planned for in the initial 1950s-era suburban build-out. The cost of installing a sidewalk during original development build-out is typically a fraction of the cost of a retrofit.

Further, in the streetcar suburbs described by Warner (1978), property lines were clearly defined by the sidewalk. However, in developments without sidewalks, property owners often claim the public right-of-way as part of their private landscaping. Retrofitting these areas requires evaluating legal easements and facing social resistance from residents who may view public sidewalk installation as an intrusion into their private property.

2.3.4 Historic Policy Shapes Equity Gaps Today

The physical fragmentation of the sidewalk network is not solely a byproduct of changing transportation technology, but also a reflection of intentional policy decisions that institutionalized location-based inequality. In *The Color of Law* (2017), Richard Rothstein argues that the segregation of American municipalities was not a result of individual choices, but rather of government action. A primary tool in the effort was the residential security maps made by the Home Owners Loan Corporation (HOLC), which graded neighborhoods from A “Best” to D “Hazardous.” These grades indicated an assessment of the physical condition of the neighborhood, as well as its racial composition and income level. Known as “redlining,” the federal government’s involvement in the lending practices based on these maps ensured that investment was directed away from the “Hazardous” areas and into the white periphery.

This systemic denial of capital influenced the location of public works projects, such as paved streets, drainage systems, and sidewalks. While the local and federal governments neglected urban cores, subsidies were granted for suburban buildout and national freeway systems – thus helping to usher in the car era.

Today, these areas look very different. In neighborhoods that survived poor redlining grades, sidewalks are generally in a worse state than historic white neighborhoods. However, some redlined areas have more complete sidewalk networks than midcentury peripheral developments. The challenge today is not in determining where the gaps are, but in deciding which areas get infrastructure improvements first.

2.4 Comparative Case Studies: Seattle, WA & Minneapolis, MN

This research employs a comparative case study approach to evaluate how development era, governance structure, and equity legacy interact to produce the sidewalk networks observed today. Denver serves as the anchor case because its 2022 voter approval of Initiative 307 and subsequent transition toward a publicly funded sidewalk program offer a contemporary, fully documented example of a city moving out of the private-liability model. To situate Denver’s reform within a broader policy landscape, Seattle and Minneapolis were selected as peer cities that share Denver’s core analytical conditions while bracketing it along the governance dimensions most relevant to this study.

Three criteria guided peer selection. First, each city contains a clear break between pre–1940 pre-car era development and post–1940 automobile-era expansion, allowing the era-based spatial analysis introduced in Section 2.3 to be applied consistently across all three networks. Second, each city operates under a distinct sidewalk governance and funding model, producing a meaningful policy spectrum: Denver has transitioned to a voter-approved dedicated fee, Seattle relies on periodic transportation levies while retaining property-owner liability, and Minneapolis retains property-owner liability but operationalizes public management through city-led inspections and assessed repairs. Third, each city has a documented HOLC redlining footprint, enabling a parallel analysis of how historic disinvestment patterns map onto present-day pedestrian infrastructure gaps. Comparable metropolitan scale, northern-tier climate,

and the operational challenge of seasonal maintenance further support direct comparison. *Table 1* summarizes each city’s contribution to the analytical framework that structures the remainder of this report.

Table 1. Case Study Comparative Framework

City	Development Trajectory	Governance & Funding Model	Equity Legacy	Analytical Contribution
Denver, CO	Streetcar core (pre–1940) surrounded by extensive post–1940 automobile-era expansion	Private-liability model superseded by Initiative 307 (2022); transitioning to a publicly funded annual fee	Strong correlation between HOLC C and D grades and present-day sidewalk gaps	Anchor case: documents the full reform arc from private liability to municipal management
Seattle, WA	Pre–1940 streetcar core with post-1953 northern annexations developed without sidewalks or curb-and-gutter drainage	Private liability retained; capital delivery funded through periodic voter-approved transportation levies	Pre-car infrastructure largely preceded HOLC grading; gaps concentrated in unsurveyed peripheral annexations	Levy-driven reform pathway and the “walkway” interim-infrastructure model for high-cost retrofit areas
Minneapolis, MN	Compact pre–1940 grid with limited car era expansion; gaps concentrated in industrial corridors and edges	Private liability retained, but operationalized through city-led rotating inspections and assessed repairs	Persistent gap correlation with HOLC C and D grades despite small overall network deficit	Operational variant of the property-owner model: city-managed repair delivery without transfer of legal liability; serves as the diagnostic comparison for inspection-and-bill programs

2.4.1 Case Study: Seattle, WA

Seattle is comparable to Denver in metropolitan size, continuing expansion of sidewalk gap funding, and the structural divide between its pre-car core and later car era expansion. The city’s pre–1940 footprint, served by an extensive streetcar network, produced a continuous pedestrian grid in central neighborhoods. Beginning in the 1950s, Seattle annexed large areas to the north that had developed under

unincorporated King County standards, where sidewalks, curbs, and stormwater infrastructure were not required. These annexations produced concentrated and contiguous sidewalk gaps that map cleanly onto the era-based typology developed in this report, making Seattle an effective test case for whether development era predicts network fragmentation when other governance variables are held constant.

Seattle also offers a governance contrast to the Denver model. Like most U.S. cities, Seattle retains property-owner liability for sidewalk maintenance under its municipal code. Unlike most peer cities, however, it funds new pedestrian infrastructure through periodic voter-approved transportation levies that direct capital investment using an equity index, proximity to amenities, and safety criteria (Seattle Department of Transportation, 2025). The City has also developed an interim “walkway” program that installs street-grade pedestrian paths in areas where full sidewalk construction would require prohibitively expensive drainage retrofits.

Seattle therefore represents a reform pathway distinct from Denver’s and provides a model for how cities can deliver interim pedestrian access where physical conditions make complete sidewalk construction infeasible in the near term. Section 5.3 returns to the levy mechanism in detail.

2.4.2 Case Study: Minneapolis, MN

Minneapolis serves as a peer city because, like Seattle, its development trajectory closely parallels Denver’s. Its physical scale, however, offers an informative contrast. The city was substantially built out during the pre-car era under a compact platting pattern, with limited post-1940 expansion; sidewalk gaps therefore concentrate primarily in industrial corridors, along rail and river edges, and in the small number of mid-century neighborhoods developed after the streetcar network was dismantled. This pattern allows the era-based analysis to be tested at a different urban scale: where Denver and Seattle present large, dispersed car era peripheries, Minneapolis concentrates its deficits in identifiable corridors. The city’s smaller land area, roughly one-third of Denver’s, also makes it a useful reference point for evaluating whether targeted, segment-based infill strategies remain viable in larger cities.

Minneapolis also occupies yet another position on the governance spectrum. The City retains property-owner liability for sidewalk maintenance but operationalizes public delivery through a rotating inspection program. In this program, City staff identify deficient segments, the City manages construction and procurement, and adjacent owners are assessed for the cost once work is completed (City of Minneapolis, 2020). This hybrid arrangement preserves the legal framework most U.S. cities operate under while ensuring that repairs proceed on a predictable schedule rather than awaiting individual owner action. Section 5.3 examines the tradeoffs of this approach in greater depth.

The historical and comparative material developed in this chapter is the foundation for the guide’s anchor case and diagnostic sections. The governance evolution traced in Section 2.2 supplies the legal and institutional context on which the guide draws when explaining why the private-liability model is a policy choice rather than an inevitability. The era-based typology developed in Section 2.3 frames the slider-map comparisons of

pre-car and car era development that appear in the guide's Denver narrative. The peer cities introduced in Section 2.4 populate the guide's comparative inventory of governance and funding models, where Seattle and Minneapolis appear alongside Denver as documented points along the reform spectrum.

Chapter 3: Methodology

This chapter documents the data sources, normalization steps, and spatial analysis framework used to produce the findings presented in Chapter 4. Methodology is treated as a first-class component of the deliverable because the guide is intended to be replicable: a city evaluating its own sidewalk governance should be able to reproduce the analytical steps described here using its own open data and a comparable historic-overlay workflow. The qualitative research described in Section 3.6 is treated with equal weight, as the subject-matter-expert interviews directly shaped which spatial questions were asked and how the resulting findings are interpreted in the guide.

3.1 Research Design

One of the primary objectives of this study is to determine the extent to which urban development eras serve as a predictor for sidewalk infrastructure gaps. To achieve this, we conducted a comparative spatial analysis across three peer cities: Denver, CO; Seattle, WA; and Minneapolis, MN.

Using an era-based typology, this research categorizes urban form into two broad periods: the pre-car era, defined as development before 1940, and the car era, defined as development from 1940 through 1990. The 1940 break reflects a major shift in transportation patterns, land-use regulation, and subdivision design, as cities moved away from streetcar- and pedestrian-oriented development toward automobile-oriented growth. Post-1990 development is excluded because the rise of New Urbanism and related planning reforms reintroduced many historic sidewalk-design principles, including detached sidewalks, minimum pedestrian clear widths, and stronger attention to the pedestrian realm. The pre-car era is generally associated with the streetcar suburb typology, which includes higher densities, connected street grids, and reliance on walking and transit. The car era contrasts with this pattern through automobile-oriented planning, lower densities, and zoning practices that often deprioritized pedestrian infrastructure.

To evaluate the impact of historical equity practices on the modern sidewalk network, we integrated redlining spatial data for each study city. This data, sourced from the University of Richmond's Mapping Inequality webtool, identifies the boundaries established by the HOLC in the 1930s. By overlaying the missing sidewalk segments onto these historical maps, the analysis identifies the correlation between residential security grades and the modern infrastructure gaps.

The methodology also examines the influence of historical transit on current infrastructure through a proximity buffer analysis of streetcar lines. For Denver and

Minneapolis, the study utilized documented historical rail alignments, while the Seattle transit layer was manually digitized and reconstructed based on a map of the streetcar network in 1931. We generated a ¼-mile buffer around these lines to represent the standard pedestrian catchment area for transit access. The intersection of streetcar buffers and sidewalk gaps would showcase the relationship between transit access and sidewalk construction.

3.2 Data Sources

We synthesized spatial data from multiple municipal open data portals. While each city provided a baseline for sidewalk inventory, the attributes and data structures varied significantly:

- **Denver:** Data was sourced from the Denver Open Data portal, providing a robust inventory including sidewalk presence, width, and material condition (2025) as well as parcel information (2020, updated in 2021), and Abandoned Trolley Tracks (2024).
- **Seattle:** Data sources include the Seattle Zoned Development Capacity Layers, which provided parcel-level information and sidewalk condition attributes (2016, updated 2024). The streetcar network map was created by tracing a 1931 historical streetcar routes map.
- **Minneapolis:** Data obtained from the University of Minnesota (2018) and Minneapolis Open Data (2026). Unlike Denver and Seattle, the Minneapolis dataset specifically isolated "sidewalk gaps" as a primary metric rather than condition attributes.
- **Redlining:** collected from the University of Richmond's Mapping Inequality webtool. This tool has the redlining maps for many American cities.

3.3 Data Normalization & Spatial Joining

Because the three municipalities use nonconforming datasets, the analysis required a normalization process to create a consistent comparative framework. The most significant technical challenge was the Minneapolis dataset, which lacked integrated development-year attributes. To address this, we performed a spatial join between the city's parcel data table, which contained "Year Built" attributes, and its land use layer. This allowed parcels to be classified into the designated "Pre-Car" and "Car" eras and produced a clearer timeline of Minneapolis's urban growth. One limitation is that the sidewalk datasets used in this report do not consistently distinguish sidewalk width. As a result, some segments classified as "existing sidewalk" may still require widening or reconstruction to meet current accessibility standards.

3.4 Spatial Analysis Framework

Once the data was normalized, we applied a spatial analysis. This involved era categorization, gap identification, and proximity analysis. To categorize parcels into different development eras, a binary classification (Pre-Car vs. Car) was created. This was based on the year a parcel's existing structure was built; 1940 was selected as the

break due to the changes in car ownership during the two decades (1930s and 1940s). Next, gaps were identified through mapping the "Missing" sidewalk conditions and can be seen in Figures 2, 3, and 4. Finally, the correlation between a neighborhood's distance from the central business district (CBD) and the frequency of sidewalk gaps was evaluated and showed more gaps are present in areas dominated by Car Era development.

3.5 Limitations

The primary limitation of this methodology is the variance of data reporting between cities. While Denver and Seattle track a detailed array of condition attributes, Minneapolis focuses more heavily on the binary presence or absence of a sidewalk. Consequently, this analysis prioritizes "Network Contiguity" (the presence of a gap) over "Material Condition" to maintain a standard metric across all three study areas. While this analysis considers sidewalk width where data are available, only Denver and Seattle classify segments outside the "existing sidewalk" / "missing sidewalk" binary as "inaccessible sidewalk." This distinction helps prevent those networks from appearing more complete than they are.

3.6 Qualitative Research

To ground the spatial data in real-world policy and engineering constraints, we conducted conversational interviews with several key stakeholders. These experts were selected to provide a multi-disciplinary view of gaps in pedestrian infrastructure and city approach.

- **Policy & Advocacy Perspective:** Jill Locantore, the author of Denver Deserves Sidewalks (Initiative 307) and the Executive Director of the Denver Streets Partnership. This provided direct insight into the political and legal hurdles of shifting sidewalk responsibility from private owners to the public sector, as well as context for the advocacy work that led to the initiative.
- **Engineering & Safety Perspective:** Wes Marshall, traffic engineer and CU Denver faculty member. This interview highlighted the technical and safety-related barriers to sidewalk implementation, as well as potential policy shifts and additional analytical metrics for future study.
- **Academic Perspective:** Garrett Fardon, MURP Program alumnus, who wrote his thesis on equity gaps in sidewalks and their relation to public transportation. This provided the theoretical framework for understanding how infrastructure correlates with socio-economic disparities.
- **Municipal Funding & Delivery Perspective:** Cailin Henley, Senior Transportation Planner with the Seattle Department of Transportation and lead of the city's Sidewalk Development Program. This interview clarified how Seattle's transportation levy funds new sidewalk construction and surfaced the distinction between two parallel programs: the Sidewalk Development Program, which delivers permanent, code-compliant sidewalks, and the Walkways Program, which installs lower-cost, at-grade walking surfaces where full retrofits are not yet feasible.

- **Assessment-Based Implementation Perspective:** Kristian Hemmesch, Transportation Planner with the City of Minneapolis. This interview clarified the limits of Minneapolis’s assessment-based sidewalk model. While the program creates a formal process for identifying and repairing deficient sidewalk segments, it has not filled sidewalk gaps in recent years and remains largely complaint-driven for repairs. Because new sidewalk construction must generally be assessed to abutting property owners, the system creates equity concerns and financial barriers that limit implementation. The interview also highlighted that Minneapolis is actively considering changes to this model, while separately piloting a more innovative winter maintenance program that shifts snow and ice removal to the City in high-traffic pedestrian corridors and safe routes to schools.

3.7 Synthesis of Data

The insights from these interviews were used to identify which type of spatial analyses would be most useful and how the resulting outputs should be interpreted. While the maps identify where the gaps are, the subject matter expert interviews explain why they persist and what specific policy levers in the sidewalk network reform guide are most likely to be effective.

The combined quantitative and qualitative methodology developed here is what makes the guide’s diagnostic content portable. The era-based binary classification, the redlining overlay, and the streetcar buffer are reproducible with municipal open data and the historic resources identified in Section 3.2; as well as the four diagnostic questions guide users are asked to answer about their own city – who is responsible for sidewalks, how that responsibility is funded, what authorization mechanism governs reform, and what pace of completion the current model produces – are direct extensions of the analytical structure laid out in this chapter.

Chapter 4: Results and Findings

This chapter presents the spatial analysis findings that constitute the empirical foundation of the guide. Conducted in ArcGIS Pro, the analysis examines how municipal scale, development history, and historical patterns of disinvestment shape the modern sidewalk network. Each finding feeds a different element of the deliverable: the comparative inventory establishes the scale of Denver’s sidewalk deficit against peer cities, the era-based analysis anchors the slider-map comparison of pre-car and car era neighborhoods, the redlining overlay grounds the guide’s equity discussion in measurable spatial terms, and the streetcar analysis frames the connection between historic transit investment and the present-day sidewalk gap.

Three findings frame the analysis. First, sidewalk fragmentation is a universal challenge across Denver, Seattle, and Minneapolis, but the form it takes differs by city. Second, among parcels with development-era data available, the car era (1940–1990) is the largest single era category for missing sidewalk in all three cities, though its share ranges from 35% in Denver to 74% in Seattle. Third, gap distribution correlates with both historical redlining patterns and proximity to historic streetcar lines, though the relationship is more nuanced than a single demographic or transit lens can capture. The chapter closes with a synthesis section that brings these three lenses together.

For each comparative section, we report results across three metrics: (1) the total miles of missing sidewalk, (2) missing sidewalk as a percentage of each city’s total network or gap inventory, and (3) gap density, measured as missing sidewalk miles per square mile of land area. Reporting all three is important because miles alone reflect city size, percentages reveal where each city’s gaps concentrate by category, and density reveals where gaps concentrate spatially.

4.1 Comparative Inventory

Across all three study cities, sidewalk gaps are widespread, but their character varies significantly.

Table 2 summarizes the inventory.

Table 2. Comparative Sidewalk Inventory: Denver, Seattle, & Minneapolis

Metric	Denver	Seattle	Minneapolis
Land area (sq mi)	155	84	54
Total missing sidewalk (mi)	318	846	283
Gap density (mi missing / sq mi)	2.05	10.08	5.25
Total inadequate sidewalks (mi)	1162	165	n/a
Gap density (mi inadequate /sq mi)	7.50	1.96	n/a

As *Table 2* shows, the three cities present three distinct patterns. Denver reports the lowest gap density at roughly 2 miles per square mile, despite having the largest land area (155 sq mi) of the three cities. The low density reflects Denver's geographically dispersed pattern of gaps: a traveler is likely to encounter a missing segment every few blocks, with larger concentrations within select neighborhoods rather than entire districts (*Figure 2*). This dispersion is consistent with how Denver expanded outward across the postwar period, when new subdivisions were approved across a broad land area without uniform sidewalk standards.

These figures capture only missing segments. Denver also carries more substandard-width sidewalk mileage than Seattle; comparable data was not available for Minneapolis. The equity gap is therefore wider than the gap-density numbers alone suggest.

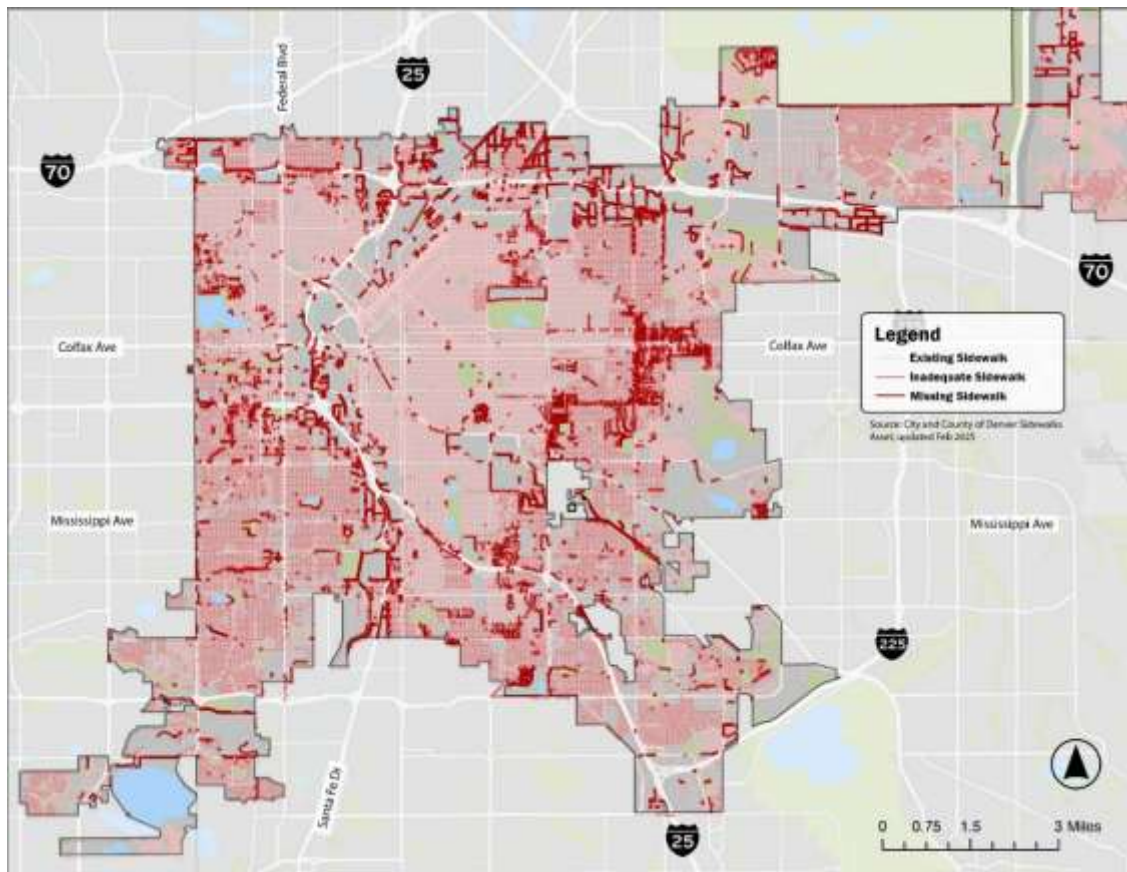


Figure 2. Sidewalk Gaps: Denver

Seattle presents a contrasting pattern. With less than half of Denver's land area but more than twice Denver's mileage of missing sidewalk, Seattle has the highest gap density of the three cities at roughly 10 miles per square mile. This concentration indicates that Seattle's gaps are not evenly distributed but cluster intensely in specific neighborhoods, particularly North Seattle, where multiple contiguous areas lack sidewalk infrastructure entirely (*Figure 3*).

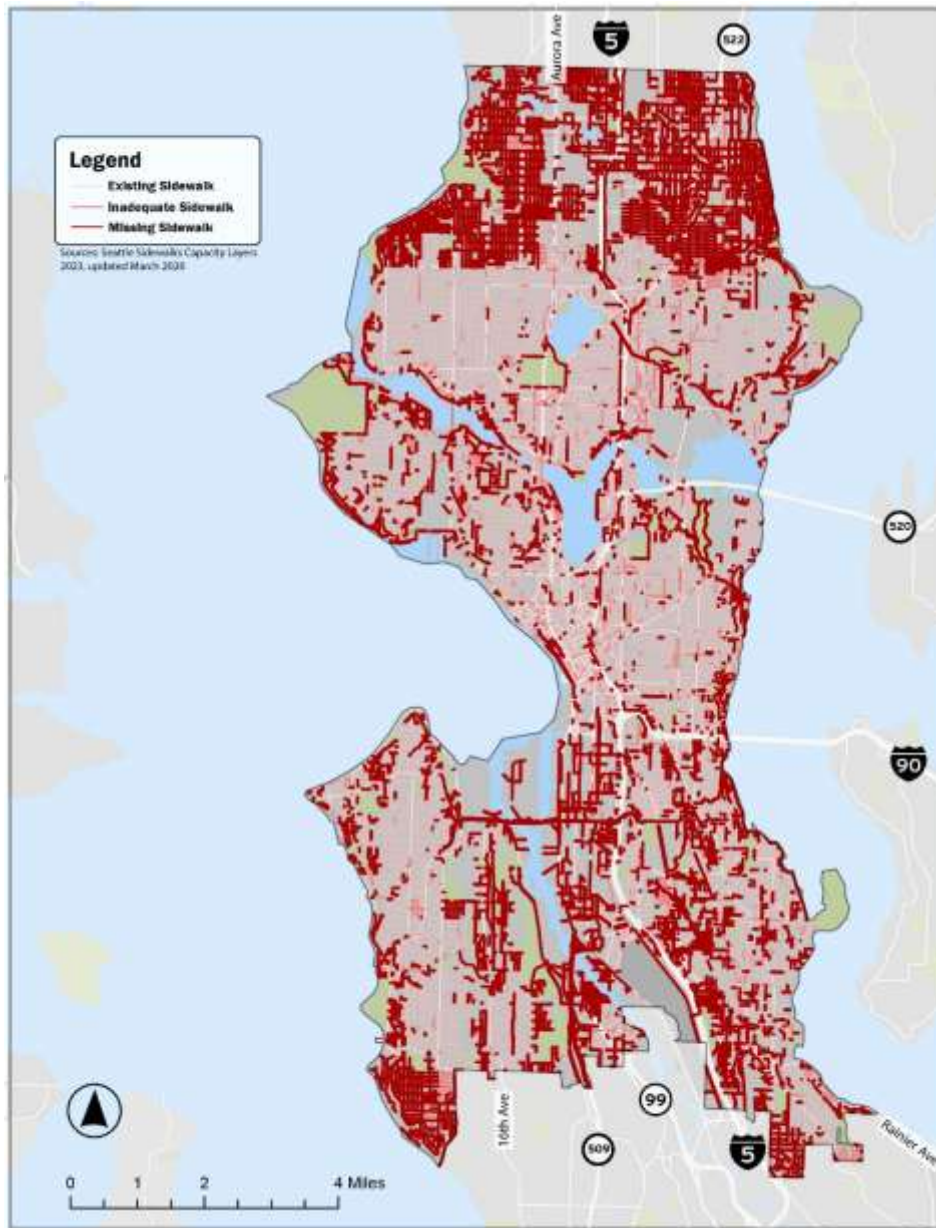


Figure 3. Sidewalk Gaps: Seattle

Minneapolis sits between the two, with a gap density of about 5 miles per square mile. Its gaps concentrate primarily along the city's periphery and in industrial areas, while central neighborhoods retain strong network continuity (*Figure 4*).

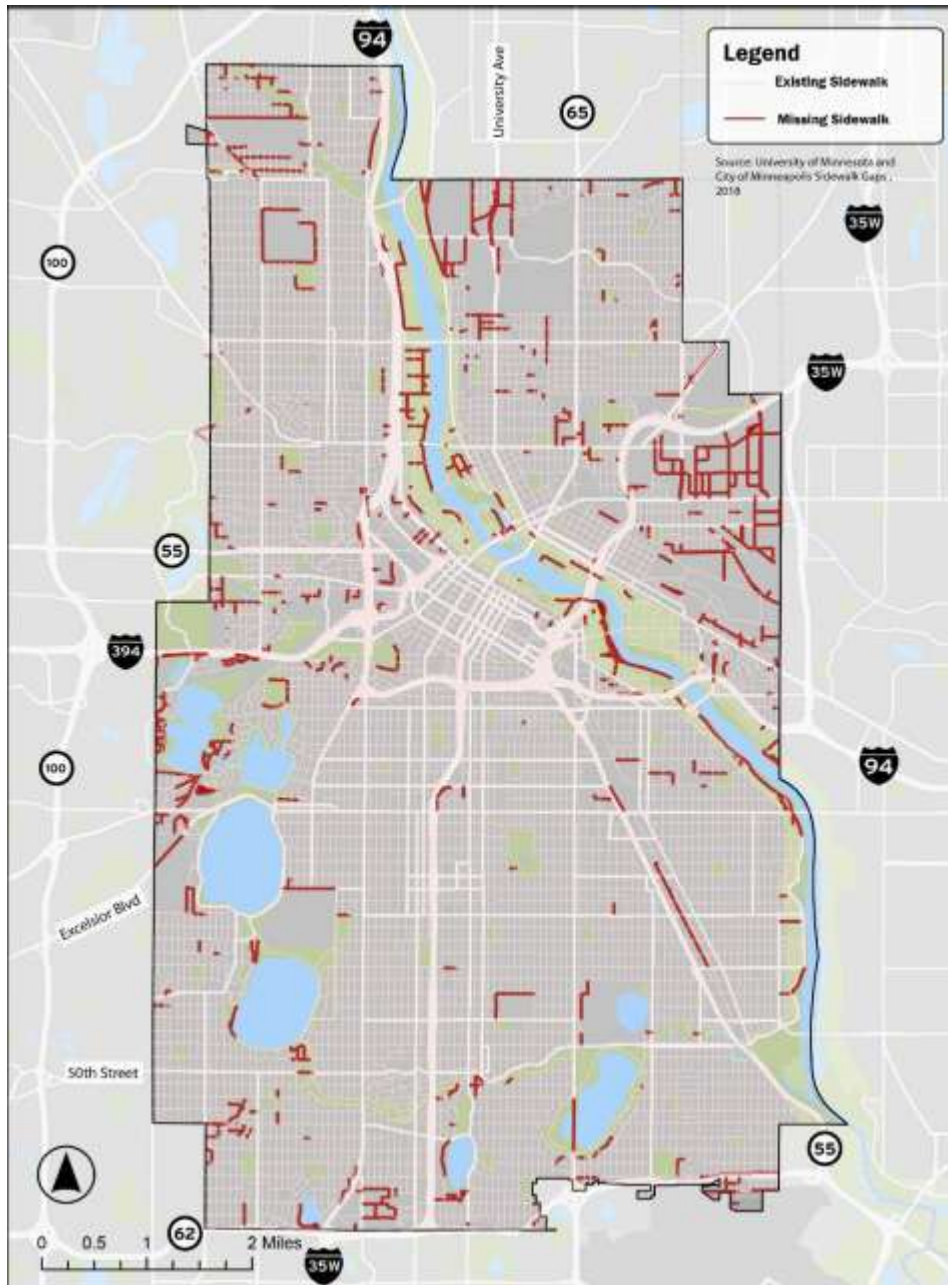


Figure 4. Sidewalk Gaps: Minneapolis

The implication of these three patterns is that no single intervention strategy fits all three cities. Denver's challenge is filling in dispersed gaps across a large land area, which requires sustained, citywide implementation rather than concentrated capital projects. Seattle's challenge is closer to localized network construction, since several neighborhoods lack basic connectivity altogether. Minneapolis falls in between: a smaller geographic footprint allows for more targeted gap-infill projects, particularly along the periphery where most gaps occur.

These differences also reflect each city's land area. Denver must manage substantially more linear feet of sidewalk than Minneapolis, which complicates maintenance and

implementation regardless of policy framework. Normalizing the data through gap density rather than raw miles makes city-to-city comparison meaningful and reveals that absolute size strongly influences the pace at which any city can complete its network.

4.2 Development Eras and Infrastructure Gaps

Among parcels with development-era data, the era in which a neighborhood was developed correlates strongly with the presence of sidewalk gaps, although the strength of that correlation varies by city. *Table 3* reports the share of missing sidewalk attributable to each era.

Table 3. Sidewalk Gaps by Selected Development Eras

Development Era	Denver (mi, % of gaps)	Seattle (mi, % of gaps)	Minneapolis (mi, % of gaps)
Pre-1940 (pre-car era)	51 (16%)	130 (15%)	50 (18%)
1940–1990 (car era)	111 (35%)	628 (74%)	116 (41%)

The era categories used here are defined in Chapter 3. As shown in *Table 3*, the car era is the largest single era category in all three cities, but the magnitude varies sharply. In Seattle, 74% of all missing sidewalks fall in car era neighborhoods, a clear supermajority. In Denver and Minneapolis, the car era accounts for 35% and 41% of gaps respectively, with pre-1940 areas adding another 16% and 18%.

The remaining share in Denver (about 50%) and Minneapolis (about 41%) corresponds to parcels not classified into either era, including post-1990 development, non-residential land, and parcels without era coding.

In Denver, the era pattern manifests as dispersed gaps across postwar neighborhoods, where larger lot sizes and greater setbacks reduced the immediate functional demand for sidewalks. *Figure 5* overlays Denver's development eras against missing sidewalks, allowing direct visual comparison of where gaps concentrate by build period.

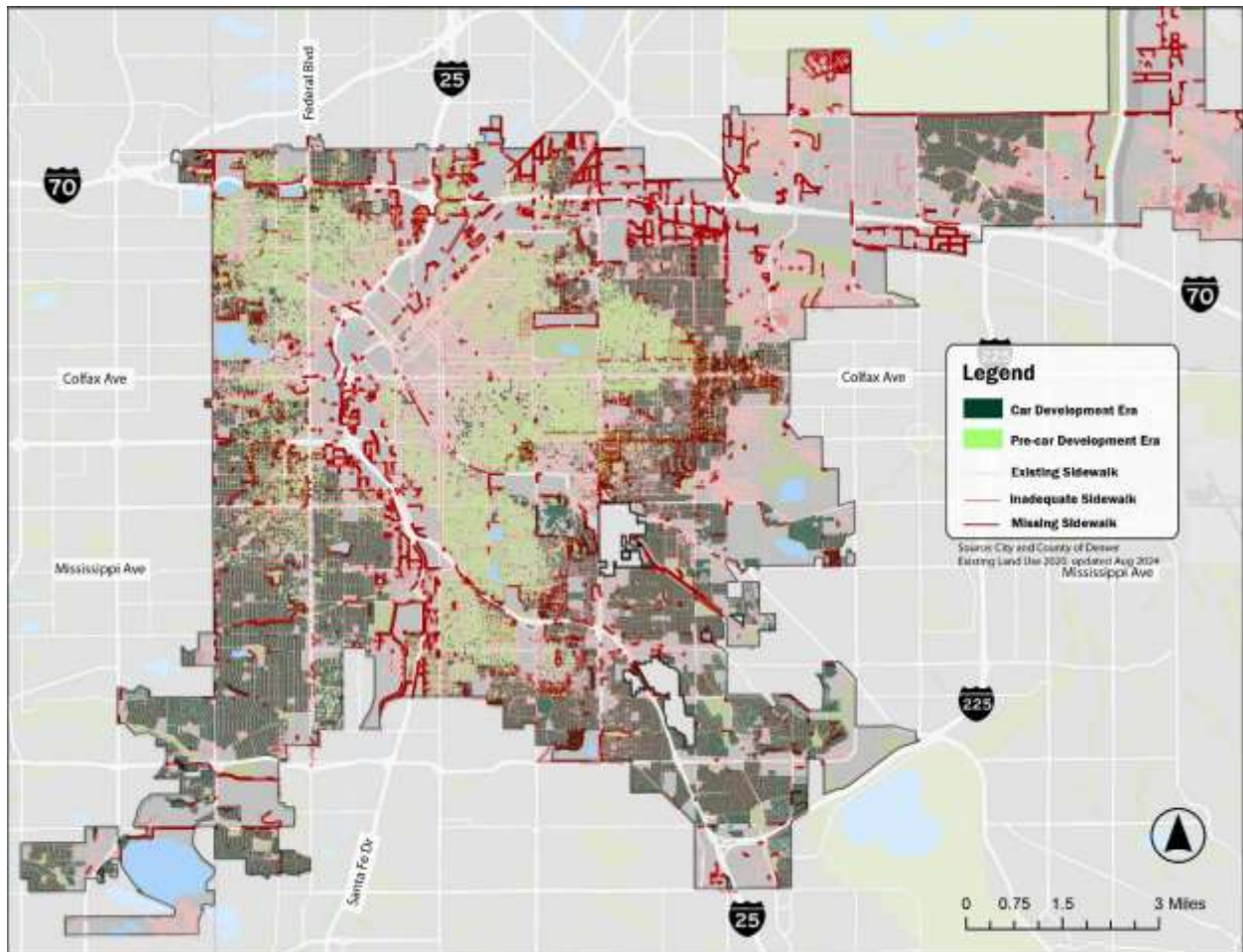


Figure 5. Development Eras & Sidewalk Gaps: Denver

In Seattle, the era pattern is more spatially concentrated. North Seattle neighborhoods developed during the car era often lack not only sidewalks but also the curb and stormwater infrastructure that pre-car era areas inherited as part of original construction. The 74% car era share in Table 3 reflects this concentration in North Seattle and other postwar annexations (*Figure 6*).

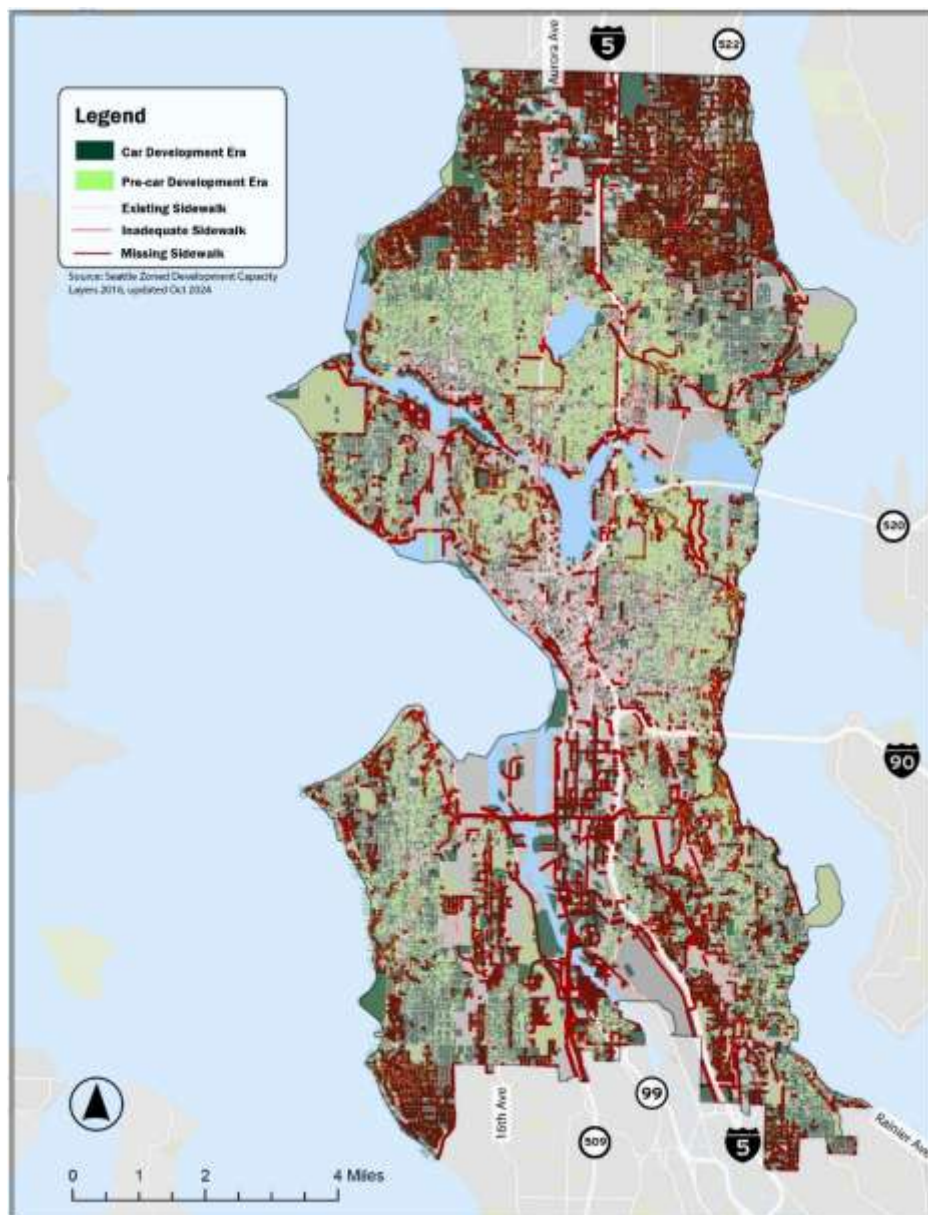


Figure 6. Development Era & Sidewalk Gaps: Seattle

Minneapolis follows the same era-based pattern but at a smaller geographic scale, with car era gaps concentrated in peripheral neighborhoods (*Figure 7*). Note, however, that data limitations in some segments may slightly underrepresent the total.



Figure 7. Development Era & Sidewalk Gaps: Minneapolis

The car era gap is best understood as a product of the regulatory environment of its period rather than any single design intent. In some cases, developers omitted sidewalks to reduce construction costs under permissive subdivision standards. In others, the absence of sidewalks reflected an aesthetic preference for a more rural or estate-style streetscape, particularly in higher-end developments. What unites these cases is that the regulatory framework of the period treated sidewalks as optional or an afterthought rather than a necessary amenity, and the result is a generation of neighborhoods built without continuous pedestrian networks.

Retrofitting these neighborhoods today is more difficult than the original installation would have been, but the difficulty is not only financial. Public perception and resident opposition can slow projects, particularly where property owners have landscaped or otherwise treated the public right-of-way as private space for decades. Design

challenges are significant where curb-and-gutter systems were never installed, since adding sidewalks requires concurrent stormwater retrofit. Tree preservation, utility relocation, and slope or grading constraints add further complications. Together, these factors explain why sidewalk gap closure remains slow in car era neighborhoods across all three cities.

4.3 Historic Equity and Redlining

To assess whether historical patterns of housing discrimination correlate with present-day sidewalk gaps, we compared each city’s missing sidewalk inventory against the 1930s Home Owners’ Loan Corporation (HOLC) maps. The HOLC maps graded urban neighborhoods on a four-tier scale from A (“best”) through B (“still desirable”), C (“definitely declining”), and D (“hazardous”), with grades C and D applied disproportionately to neighborhoods with Black, immigrant, or working-class residents. These grades guided federal mortgage insurance decisions for decades and are widely used in planning research as a proxy for historical disinvestment. Areas of each city that were not surveyed by HOLC are reported separately as a fifth, “ungraded” category. *Table 4* presents these results.

Table 4. Missing Sidewalk by HOLC Grade

HOLC grade	Denver (mi, % of total)	Seattle (mi, % of total)	Minneapolis (mi, % of total)
A – Best	22 (7%)	28 (3%)	31 (11%)
B – Still Desirable	19 (6%)	103 (12%)	36 (13%)
C – Definitely Declining	41 (13%)	129 (15%)	54 (19%)
D – Hazardous	42 (13%)	62 (7%)	53 (19%)
Ungraded	194 (61%)	526 (62%)	110 (39%)

In Denver and Minneapolis, areas graded C and D show notably higher rates of missing sidewalks than areas graded A or B. In Denver, C and D areas together account for 26% of total gaps (41 + 42 miles) while A and B areas account for only 13% (22 + 19 miles). These patterns are similar for Minneapolis. This is illustrated in *Figure 8* and *Figure 9*. In these cities, the spatial analysis highlights a clear trend where the most vulnerable populations were left with fragmented pedestrian networks, creating a barrier to economic and social mobility that still exists today.

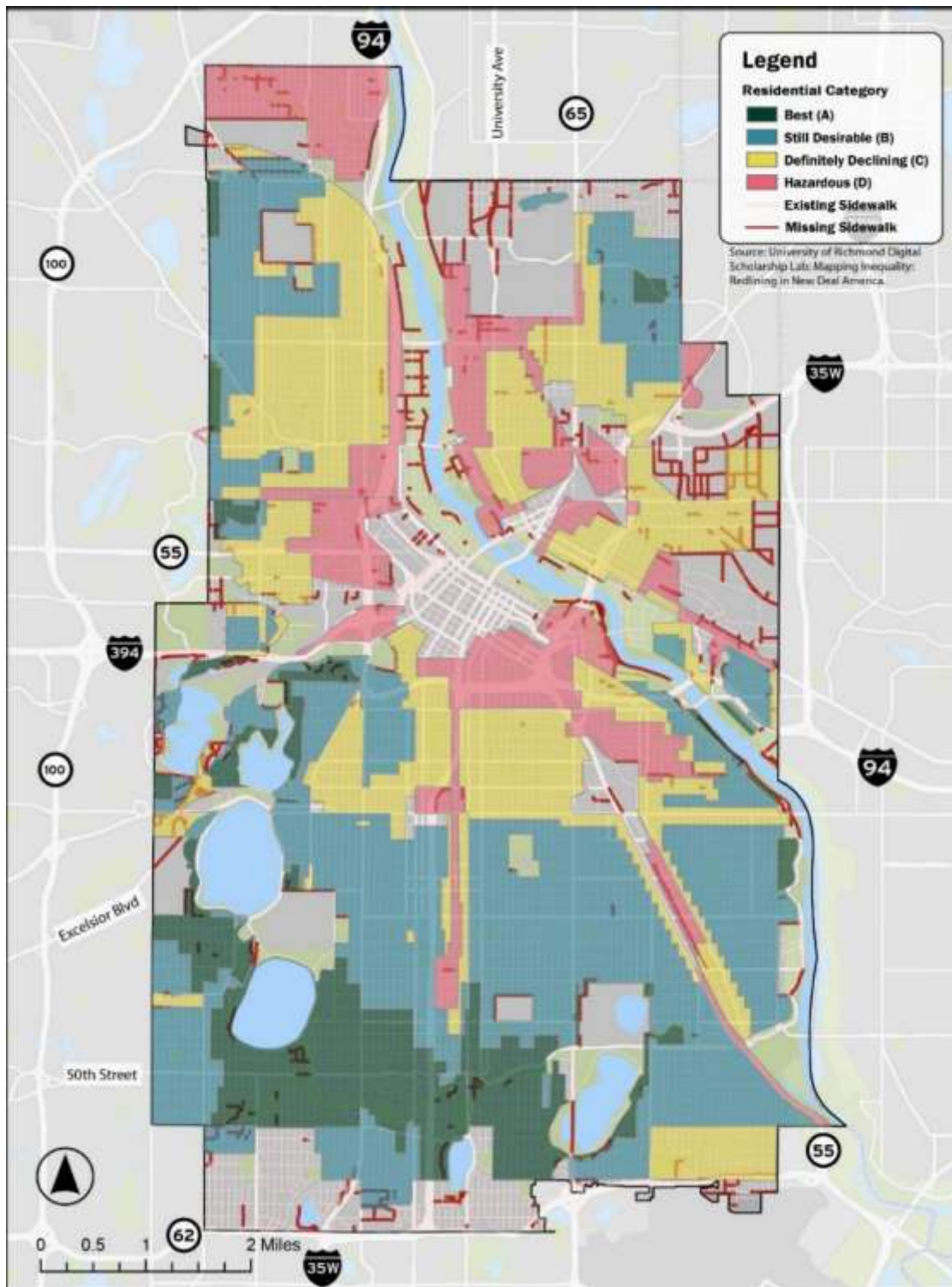


Figure 8. Redlining: Minneapolis

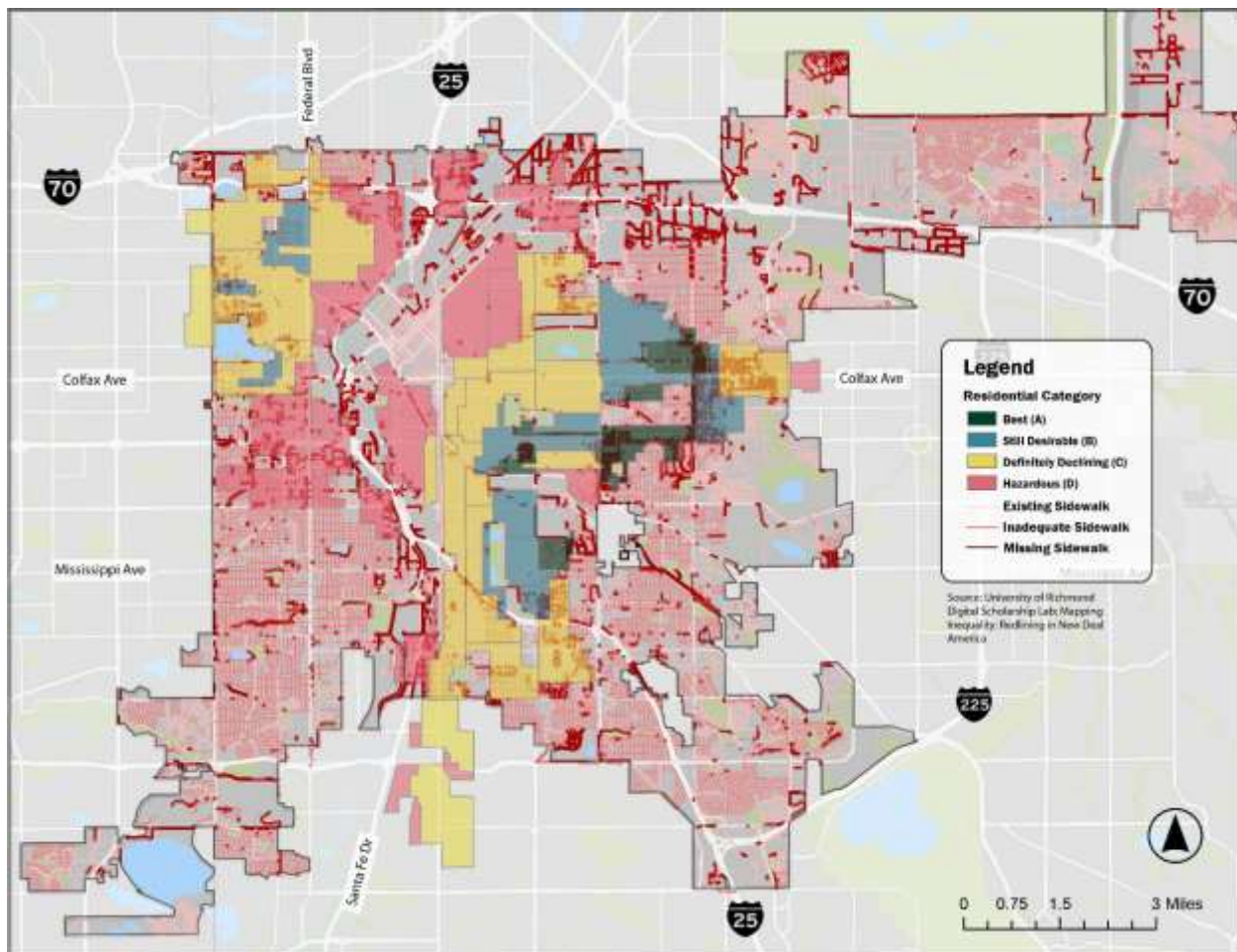


Figure 9. Redlining: Denver

Seattle presents a different story. As *Figure 10* shows, Grade D areas in Seattle account for only 7% of total gaps, less than Grade B (12%) or Grade C (15%). Two factors likely contribute to this. First, Seattle's central and downtown neighborhoods, which include several Grade C and D areas, are among the city's oldest pre-1940 districts and would have been built with sidewalks as part of pre-car era construction regardless of later HOLC grading. Second, the largest concentrations of missing sidewalks in Seattle fall in North Seattle, much of which was undeveloped at the time the HOLC maps were drawn and was later built out as car era residential. These areas appear in our analysis as ungraded rather than as a specific HOLC grade. Distinguishing the older-but-redlined pattern from the never-surveyed-and-car-era pattern is essential to interpreting Seattle's results without overstating the role of redlining in that city specifically.

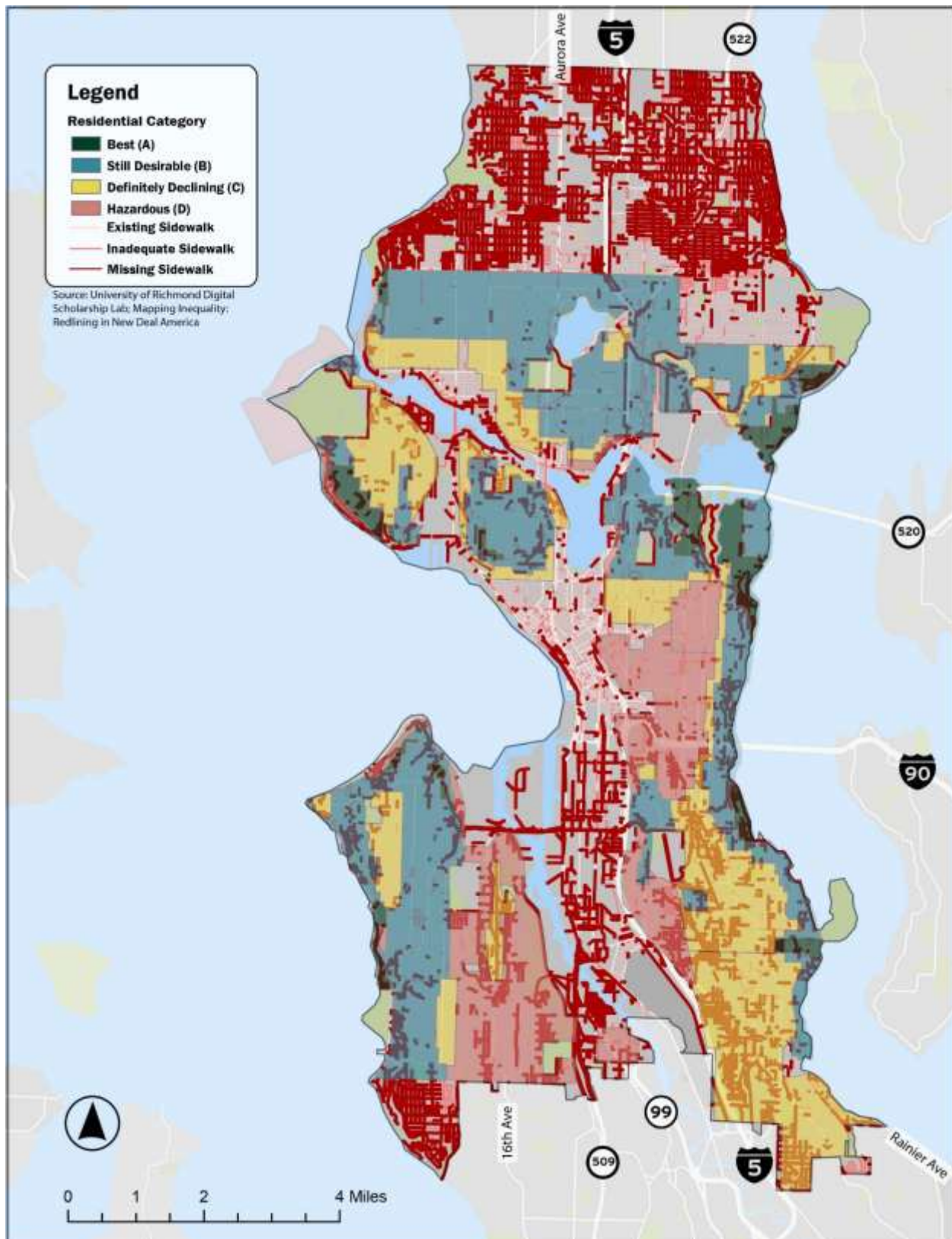


Figure 10. Redlining: Seattle

A finding visible in all three cities, though strongest in Denver and Seattle, is that ungraded areas account for the largest single-category share of missing sidewalk: 61% in Denver, 62% in Seattle, and 39% in Minneapolis. These non-surveyed areas were typically undeveloped, agricultural, or nonresidential at the time the maps were drawn and later transitioned into car era suburban development. They inherited neither the pre-car era sidewalk networks of the urban core nor the scrutiny of the redlining maps. The result is a second, distinct equity layer that any prioritization framework focused exclusively on historically redlined areas will miss.

4.4 The Transit Oriented Legacy

The third spatial lens examines whether historic streetcar service correlates with present-day sidewalk completeness. We mapped each city's historic streetcar lines and applied a quarter-mile buffer (approximately a five-minute walk) around each line to capture the area where streetcar era pedestrian demand would have been highest. *Table 5* reports the results.

Table 5. Missing Sidewalks Within and Outside the Historic Streetcar Buffer

Buffer location	Denver (mi, % of gaps)	Seattle (mi, % of gaps)	Minneapolis (mi, % of gaps)
Within ¼ mi of historic streetcar	207 (65%)	253 (30%)	188 (66%)
Outside ¼ mi of historic streetcar	111 (35%)	593 (70%)	96 (34%)

This highlights that while pre-car areas were built to support transit-oriented living, the actual construction of sidewalks sometimes lagged behind the laying of tracks. During the pre-car era, sidewalk installation still frequently fell under the purview of private property owners or local developers rather than a centralized municipal mandate, except within specialized districts. In lower-income neighborhoods, the high cost of construction and assessments meant that some property owners deferred sidewalk installation. This resulted in a fragmented landscape where high-quality transit was mixed with low-quality pedestrian infrastructure. Within a quarter mile of streetcar lines, we determined that, while there was likely demand for sidewalks, construction did not always follow.

In Denver, the East Colfax neighborhood serves as one example of this lack of sidewalk development, as seen in *Figure 11*. While it was anchored by a major streetcar line, the neighborhood's sidewalk network reflects inconsistent investment. The presence of transit increased land value and demand for pedestrian access, yet the physical infrastructure remains incomplete.

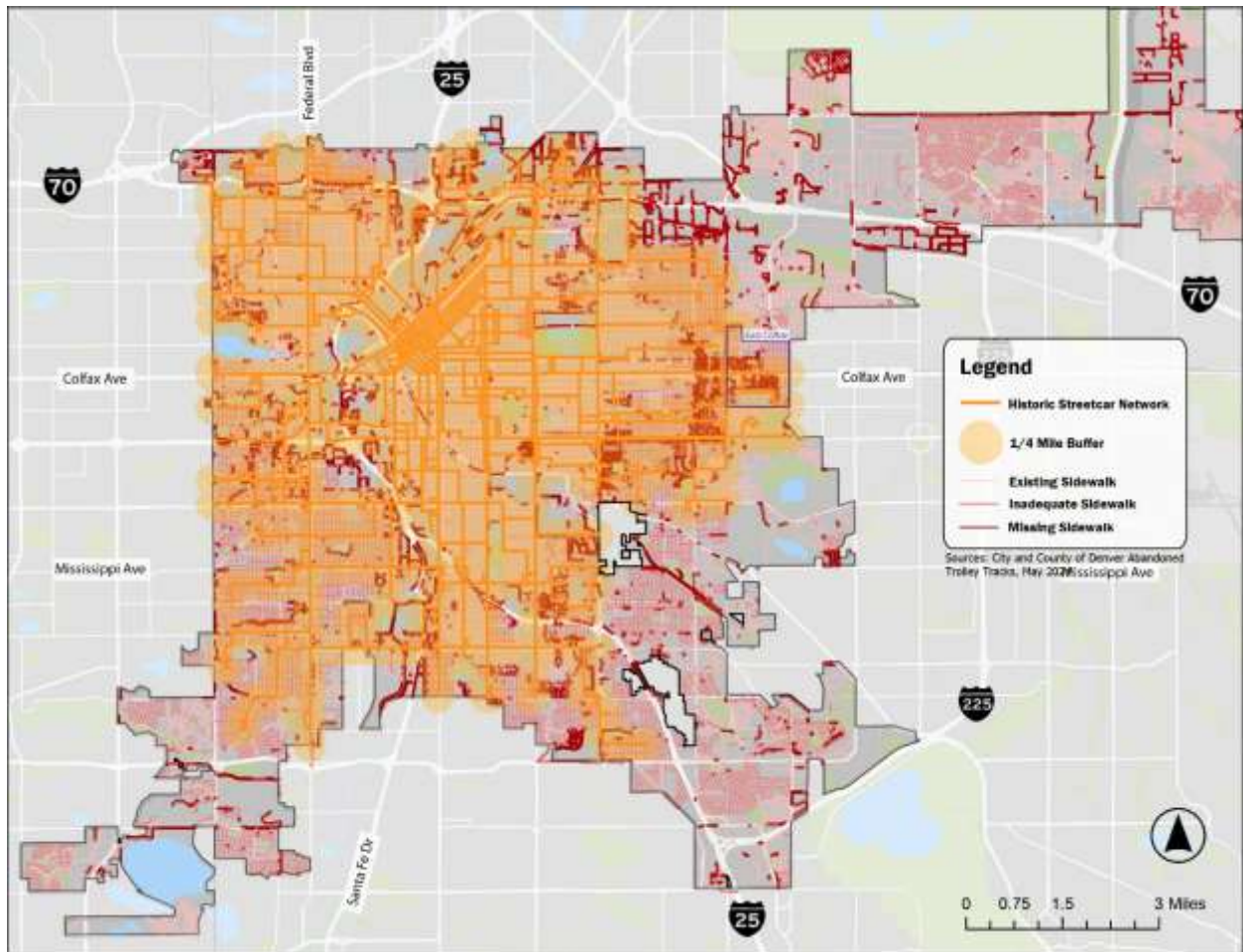


Figure 11. Historic Streetcar Lines: Denver

Minneapolis presents a similar narrative where missing sidewalk segments are found both within and immediately adjacent to the historic service area, as seen in *Figure 12*. These gaps suggest that even in transit-rich environments, residents were sometimes made to walk in the street or on unpaved paths to reach transit stops, indicating that access was a secondary priority to the transit line itself.

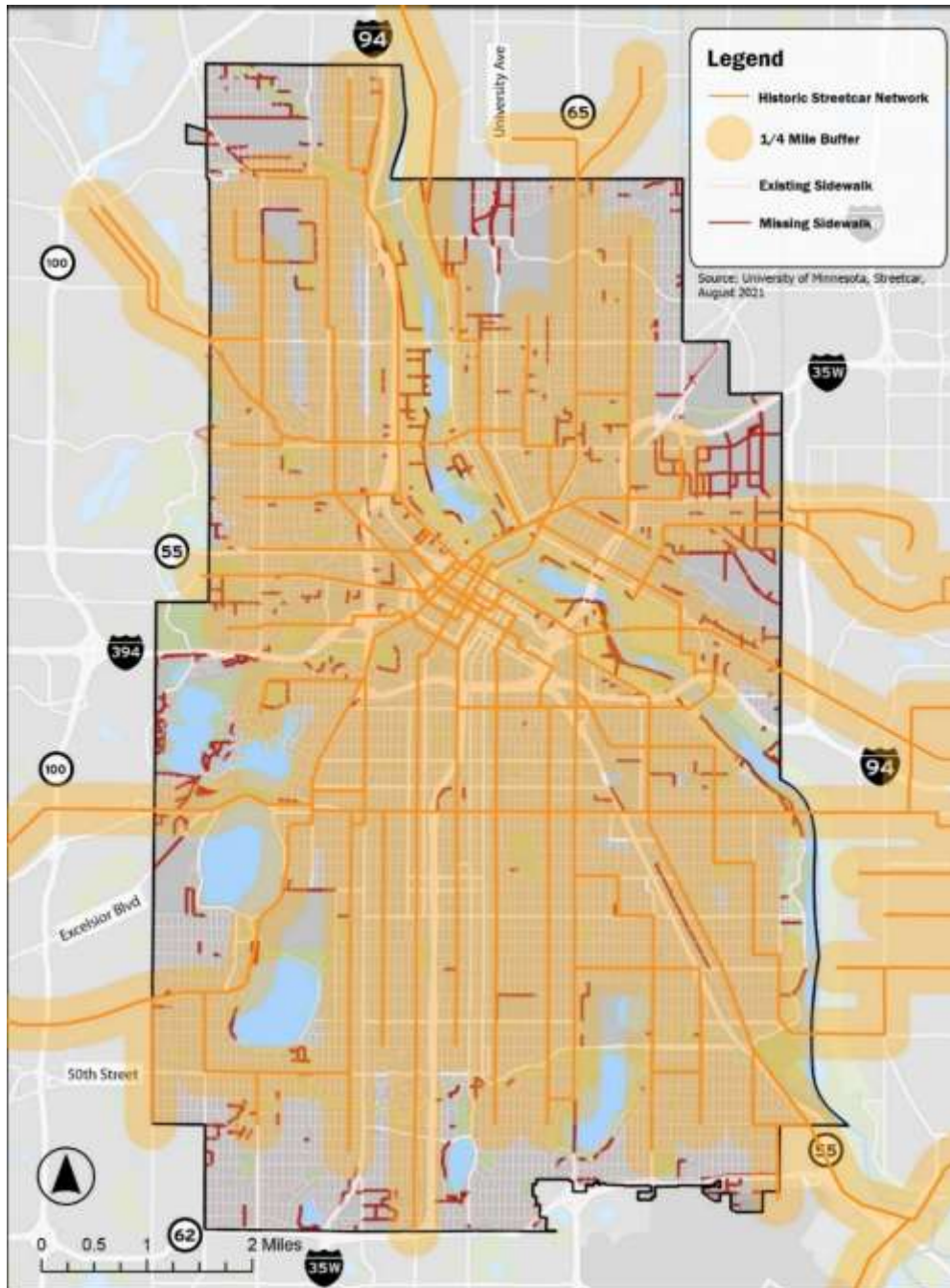


Figure 12. Historic Streetcar Lines: Minneapolis

A significant limitation of this analysis is the reliance on historic streetcar route maps rather than specific stop-level data, which is largely unavailable. It is possible that sidewalk infrastructure was more concentrated around specific stops and intersections, with gaps widening in the mid-block segments. Without more granular data, we acknowledge that these fragmentation patterns might be more nuanced at the block level. Regardless of these nuances, the data confirms that proximity to transit did not automatically increase municipal investment in pedestrian safety. In lower-income areas, the burden of connectivity was often placed on those least able to afford it.

In Seattle, as shown in *Figure 13*, the streetcar network highlights a distinct geographic divide. The system never extended into the northernmost neighborhoods that later saw rapid development post-1940. This confirms those areas were entirely built for automobile use and explains why sidewalk gaps are so much higher there than elsewhere. Because transit efficiency is inherently tied to a high-quality sidewalk network, current investments must prioritize closing these historical gaps to ensure that access is equitable and physically accessible for all users, regardless of the era in which their neighborhood was built.

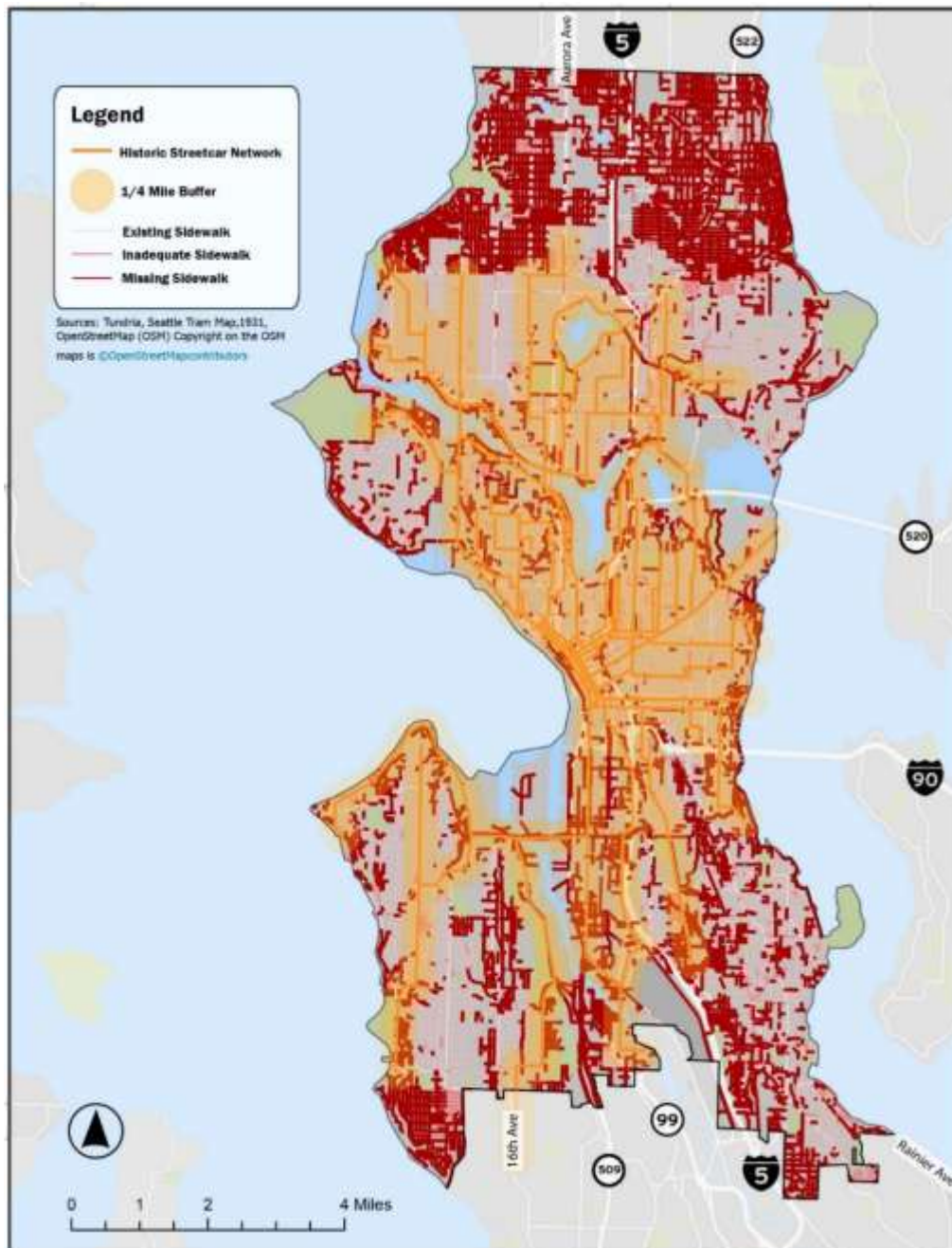


Figure 13. Historic Streetcar Lines: Seattle

Together, these findings show how the guide translates the report's spatial analysis into a practical diagnostic framework for sidewalk reform. The era-based and redlining maps appear as side-by-side slider comparisons, allowing a viewer to see directly how development era and historic disinvestment map onto present-day gaps. The streetcar buffer analysis frames the guide's discussion of how transit investment without coordinated sidewalk investment can replicate, rather than resolve, historic inequities in pedestrian access.

Chapter 5: Discussion and Policy Analysis

This chapter interprets the spatial findings presented in Chapter 4 through three policy lenses: path dependency, the shifting legal foundation of sidewalk liability, and comparative governance models across U.S. municipalities. It closes with a discussion of equity as an implementation lens that cuts across all three. The lenses structure the project's final deliverable guide, *Mind the Gap: Lessons from Denver to Rebuild Your Sidewalk Network*. The website format was chosen because the guide is intended to function as a practical planning tool, not only a written report. It allows users to move between maps, historical context, and reform pathways.

Each section of the website corresponds to a portion of the analysis developed in this report. Path dependency explains why the guide opens with a Denver narrative that begins in 1881; the legal foundation analysis supports the guide's diagnostic framing of the private-liability model as a deliberate policy choice; the comparative governance typology populates the guide's reform pathway inventory; and the equity analysis sets the conditions by which those pathways are evaluated.

5.1 Path Dependency and the Era-Based Gap

The era-based pattern documented in Section 4.2 is not coincidental. The concentration of sidewalk gaps in car era neighborhoods – most pronounced in Seattle but visible across all three cities – illustrates what Dunham-Jones and Williamson (2011) describe as path dependency: the condition in which decisions made early in a system's development constrain the options available later, often regardless of whether those decisions still serve current needs. In the case of sidewalk infrastructure, the decision to omit or under-build sidewalks during a neighborhood's initial development created a built environment that is now substantially more difficult and expensive to retrofit than it would have been to construct correctly from the start.

The pre-car era (pre-1940) neighborhoods documented across all three cities inherited near-complete sidewalk networks because the built form of that period demanded them. As Warner (1978) describes, narrow lots, shallow setbacks, and commercial corridors within walking distance of residential blocks meant that foot traffic was the primary mode of daily circulation, not an optional one. Sidewalks were installed as a default feature of the street cross-section, often alongside curb-and-gutter systems that integrated drainage and pedestrian access into a single design. Nearly a century later, that legacy remains visible: pre-1940 areas in Denver, Seattle, and Minneapolis are the ones that still maintain the highest network continuity in our spatial data.

The car era (1940–1990) produced a fundamentally different built form. As Melosi (2004) describes, the transition from the walking city to the automobile city reorganized the relationship between land use, transportation, and public space. Larger lot sizes, wider setbacks, cul-de-sac patterns, and the separation of residential and commercial uses all reduced the functional demand for pedestrian infrastructure. Subdivision regulations of the period treated sidewalks as optional, and developers typically omitted them to reduce construction costs.

The cost asymmetry created by these decisions is the core mechanism of path dependency. Installing sidewalks during initial development is inexpensive, since the right-of-way is open, drainage is being graded, and curb-and-gutter is already part of the street cross-section. Retrofitting a car era neighborhood today is several times more expensive, because the original construction often skipped curb-and-gutter altogether, established drainage patterns must now be reworked, public right-of-way may have been informally claimed by adjacent property owners, and resident expectations about the streetscape have hardened over decades. Chapter 4 documents the spatial extent of this challenge, most starkly in Seattle, where 74% of all sidewalk gaps fall in car era neighborhoods. The policy implication is that uniform implementation strategies will not work. A pre-car era neighborhood that needs repair and widening presents fundamentally different engineering, cost, and community engagement challenges than a car era neighborhood that has never had sidewalks at all. Some peer cities have responded creatively to this asymmetry. Seattle's walkway program (*Figure 14*) installs lower-cost, street-grade pedestrian paths in areas where full sidewalk construction would require prohibitively expensive drainage reconfiguration (Seattle Department of Transportation, 2025), providing interim accessibility while longer-term solutions are planned.



Figure 14. A Seattle painted walkway with wheel stop delineators and flex-posts.
Image retrieved from <https://www.seattle.gov/transportation/projects-and-programs/programs/pedestrian-program/new-sidewalks-and-walkways-program>

5.2 Shifting the Legal Foundation: From Private Liability to Public Responsibility

As discussed in Chapter 2, the legal foundation for sidewalk governance in the United States rests on a tension between two principles: the municipality's police power to require abutting property owners to build and maintain sidewalks, and the municipality's tort liability for injuries that occur on public infrastructure it has a duty to keep safe (McQuillin, 2025). For most of the twentieth century, cities resolved this tension by placing the maintenance obligation on adjacent property owners while the liability exposure sat in a complex middle ground: under most state law, where a property owner has been assigned the duty of maintenance, that owner also carries primary liability for injuries caused by their failure to maintain, but the city retains residual responsibility wherever it has actual or constructive notice of a hazardous condition and fails to act. The result was a system in which the physical form of the sidewalk network became a product of who owned what, when they built it, and whether anyone enforced the rules.

The passage of the Americans with Disabilities Act (ADA) in 1990 and subsequent rulings such as *Barden v. City of Sacramento* (2002) added a federal enforcement layer that made the private-liability model increasingly difficult to sustain in its passive form. Cities could no longer defer entirely to abutting owners; once they had notice of accessibility deficiencies in the public right-of-way, they were obligated to act. The *Willits v. City of Los Angeles* settlement of 2016, which required \$1.37 billion in repairs, demonstrated in stark financial terms what happens when a major city defers reform too long.

5.2.1 Denver's Reform Trajectory

Denver's experience illustrates both the durability of the private-liability model and the difficulty of replacing it. The 1898 *Compiled Ordinances and Charter* (City and County of Denver, 1894), describes a remarkably functional governance framework in the form of Sidewalk Districts. In these districts, an official sidewalk contractor was selected by competitive bid, City Engineer inspection and certification took place, and there was an enforcement mechanism that converted unpaid assessments directly into property tax liens. Width standards in the original code – twelve feet in front of commercial properties, five feet in front of residential and vacant lots – met or exceeded what modern ADA standards require. Outside of these Sidewalk Districts, a majority of property owners on each block could petition the city to build out sidewalks on their block if one did not already exist. In either case, the cost of construction was assessed against the abutting property in proportion to its frontage and collected via a property tax lien. The districts coordinated construction; they did not pool cost.

This system built hundreds of miles of sidewalks across early Denver. What undermined it was not neglect but expansion. As the city grew rapidly outward in the postwar era, the Sidewalk District system was phased out and the institutional machinery – Board initiative, competitive bidding, City Engineer inspection, uniform standards – was dismantled to accommodate automobile-oriented growth. The abutter-pays cost structure remained in place while the apparatus that had enforced network continuity disappeared. New subdivisions were approved with narrow sidewalks attached to the curb or, in many cases, with no sidewalks at all.

By 1988, a City Council panel sought a complete rewrite of sidewalk management rules, and the city effectively retreated from active sidewalk management. The vacuum persisted through the 2000s and into the 2010s. National research from Bridging the Gap (2012) documented that low-income neighborhoods were significantly less likely to have sidewalks, a pattern mirrored in Denver's redlined neighborhoods. By 2015, the "Rethinking Denver Sidewalk Policy" brief (WalkDenver & Mile High Connects, 2015b) reported that 47% of streets in low-income areas lacked adequate sidewalks. The brief also noted that a fully developed 2002 Right-of-Way Sidewalk Initiative (City and County of Denver, 2002), which would have established a fee of \$6.60 per square foot and an annual maintenance charge of approximately 20 cents per square foot, was never adopted.

In 2016, Denver City Council conducted a formal review of sidewalk programs in eight peer cities, including Boston, Milwaukee, and Seattle. After consulting with the City Attorney's Office, Council concluded that responsibility for sidewalk repair should remain with property owners "to protect the city from liability and to align with existing city ordinance" (City Auditor, 2020). This was the clearest institutional opportunity to shift governance before Initiative 307, and the city declined to take it due to liability concerns.

The 2017 Denver Moves: Pedestrians plan (City and County of Denver, 2019) quantified the full scale of the gap at \$1.1 billion. The \$47.7 million from the 2017 Elevate Denver Bond (City and County of Denver, 2017) covered only a fraction of the need. When the city launched its Neighborhood Sidewalk Repair Program, it stalled in its first neighborhood. After 1,100 inspections over three years, the pace was so slow that the City's own estimates projected it would take 400 years to repair the network citywide. A 2020 City Audit formally labeled the program "flawed" and confirmed that it placed an undue burden on residents.

The shift ultimately came not from City Hall but from the electorate. In 2022, the Denver Deserves Sidewalks campaign collected nearly 20,000 signatures to place a new funding model on the ballot. That November, voters approved Initiative 307, transferring responsibility for sidewalk repair and construction from abutting property owners to the City, funded by a dedicated annual fee on abutting property owners. Following Council refinements that adjusted the fee structure, the city began collecting fees in January 2025 under a nine-year master plan to build and repair a complete, ADA-compliant sidewalk network.

5.2.2 What Initiative 307 Looks Like Today

The program developed from Initiative 307 rests on three structural elements. First, the City has assumed legal responsibility for sidewalk construction and repair across all parcel types, replacing the abutting-owner obligation with a public ownership framework. Second, the program is funded by a dedicated annual fee billed to each parcel rather than by general fund appropriation, which protects sidewalk funding from competition with other budget priorities. Third, the fee is structured as a flat rate for most residential lots up to a defined frontage threshold, with a per-foot adjustment that scales the charge for properties with longer frontages. Income-qualified households are eligible for rebates that further adjust the distributional impact.

5.2.3 What Denver's Trajectory Demonstrates

The significance of Denver's trajectory extends beyond Denver itself. Although we are still awaiting conclusive results of success, Initiative 307 demonstrates that the property-owner model can be replaced. The Denver experience also shows that the replacement process is neither quick nor simple. More than two decades elapsed between the first serious policy proposals and the voter-approved reform. The advocacy pathway required sustained public engagement, coalition-building across disability rights and pedestrian safety communities, and a series of visible government failures that created the political conditions for change. Denver's reform arc is instructive in both its eventual success and the lessons learned through trial and error along the way.

5.3 Comparative Insights Across Governance Models

Municipalities use a wide range of structures to govern and fund their sidewalk networks, but those structures fall into two structurally distinct groups. The first retains the legal model in which adjoining property owners remain liable for sidewalk repair, varying only in how the city enforces that liability. The second transfers cost, legal liability, or both onto a public or collective mechanism. The distinction matters because the two groups produce different outcomes, require different authorization pathways, and place different demands on staff and budget. Table 6 organizes the comparative inventory we compiled from municipal codes, policy briefs, the Denver Auditor's 2020 review of peer programs (O'Brien, 2020), and the 2019 Denver City Council Legislative Services research report on sidewalk funding mechanisms (Griffin, 2019).

Table 6. Typology of Sidewalk Governance and Funding Models

Model type	Mechanism	Example cities	Authorization pathway	Key tradeoff
Tier 1: Property-owner liability retained				
Complaint-driven	City responds to complaints; owner pays or city assesses cost	Default in most U.S. cities	None required (status quo)	Lowest administrative burden; slowest pace; most fragmented outcomes
Inspection-and-bill	City inspects on a rotating cycle, makes repair, assesses owner	Minneapolis (MN); Denver pre-Initiative 307 (NSRP, 2017–2020)	Council action	Faster than complaint-driven on paper; in practice depends on staffing and political will; preserves cost burden on owners
Shared-cost / cost-share	City pays a portion of owner repair costs	Boulder (CO), Flint (MI), Richardson (TX)	Council action	Reduces owner share but participation is gated by ability to pay
Point-of-sale	Repair required at property transfer	Piedmont (CA); formerly Pasadena (CA), Grand Rapids (MI)	Council action	Captures some repair via market turnover; raises legal and political opposition; uneven by neighborhood

Model type	Mechanism	Example cities	Authorization pathway	Key tradeoff
Tier 2: Cost, legal liability, or both transferred to a public or collective mechanism				
Public ownership (general fund)	City assumes full liability; funded from the general budget	Boston (MA), Maitland (FL), Spanish Fork (UT)	Council action	Simple administration; funding competes annually with other priorities
Public ownership (partial)	City assumes maintenance; owner remains responsible for new construction	Lincoln (NE)	Voter mandate (1990 and 1992)	Reduces owner burden without full fiscal exposure; gap remains for new construction
Utility fee	Flat monthly charge billed with utilities	Payette (ID), Westminster (CO), Englewood (CO), Baker City (OR)	Council action	Predictable revenue; flat fees can be regressive
Dedicated annual fee	Parcel-based fee outside utility billing	Denver (CO), Initiative 307	Voter mandate	Strong revenue protection; long advocacy timeline; flat fee requires equity adjustments
Property tax millage	Dedicated millage rate	Ann Arbor (MI)	Voter mandate, periodic renewal	Substantial revenue; subject to renewal cycles
Parcel tax	Per-parcel or per-square-foot tax	Berkeley (CA), Measure FF	Voter mandate (simple majority under CA Prop 13)	Broader base than millage where state law restricts millage

Model type	Mechanism	Example cities	Authorization pathway	Key tradeoff
Sidewalk improvement district	Geographic districts with fees scaled by property classification	Ithaca (NY)	Council action, subject to permissive referendum	Supports equity scaling; adds administrative complexity
Combined transportation funding	Voter-approved levy or millage funding streets and sidewalks together	Seattle (WA), East Grand Rapids (MI), Cheney (WA)	Voter mandate (millage); council action (utility-based, as in Cheney)	Larger revenue pool; sidewalks compete with street priorities; Seattle funds new construction only, retains private liability for maintenance

Two patterns become clear once the typology is organized this way. The first is that the upper tier – programs that retain property-owner liability – describes the status quo in nearly every American city. The second is that operational reform within the upper tier does not, on its own, produce a meaningfully different outcome from the property-owner model itself. Denver’s Neighborhood Sidewalk Repair Program, the inspection-and-bill model the City operated from 2017 through 2020, illustrates the point directly. The 2020 City Auditor’s report found that the program would take more than fifty years to complete at its existing pace, that roughly eighty percent of inspected sidewalks needed repair, and that completed repairs frequently failed to meet ADA cross-slope and elevation standards (O’Brien, 2020). The program operated essentially the way Minneapolis’s program operates today: rotating inspections, city-managed repair, owner assessment. The Denver experience suggests that the operational efficiencies of an inspection-and-bill program do not, on their own, overcome the structural limitations of the model it sits inside.

Within the upper tier, inspection-and-bill programs preserve the legal framework of property-owner responsibility while attempting to deliver repairs on a predictable schedule. The city identifies noncompliant sidewalks through a rotating inspection cycle, makes the repair, and assesses the owner. Minneapolis is the most-cited contemporary example, but the same approach operates in some form in many of the cities the Denver Council reviewed in 2016, including Baltimore and Milwaukee. Shared-cost and point-of-sale programs adjust who pays the bill rather than who owes the duty: cost-share programs in Boulder, Flint, and Richardson reduce the owner’s share, while point-of-sale provisions in Piedmont (and formerly in Pasadena and Grand Rapids) require sidewalk repair at property transfer. None of these mechanisms removes the underlying liability, and none addresses sidewalks on parcels that do not change hands, are not

the subject of a complaint, or are not within an active inspection region in a given year. Cities that have operated within this tier for decades typically report the same pattern: visible deficits accumulate faster than the program resolves them.

The lower tier moves cost, legal liability, or both onto a public mechanism. **Public-ownership** models, used in Boston, Maitland, and Spanish Fork, fund construction and maintenance through the general budget. The advantage is administrative simplicity; the cost is that sidewalk funding competes annually with every other priority and can be cut during downturns. Lincoln, Nebraska illustrates a partial transfer that voters approved in 1990 and 1992: the City assumed maintenance responsibility while owners remain responsible for new construction. The Lincoln model is useful for cities that want to address visible deferred maintenance without taking on full capital obligation in one step.

Utility-style fees create a protected revenue stream by attaching sidewalk charges to existing billing infrastructure. Payette (\$1 per month residential, \$2 commercial) and Westminster (\$6 per month) take this approach. Englewood operates an opt-in “concrete utility” at approximately twelve cents per square foot of sidewalk, with ninety-five percent voluntary participation; the high uptake reflects the alternative, since if a homeowner declines the utility and their sidewalk fails inspection, the City makes the repair and places a lien on the property. Baker City charges a similar fee but directs revenue into grants for homeowner repairs, creating a hybrid incentive structure.

Dedicated annual fees, the model Denver adopted under Initiative 307, are structurally similar to utility fees but typically operate at larger scale and, in Denver’s case, were voter-mandated. Dedicated fees offer revenue predictability and political durability because they do not require periodic reauthorization. Flat-fee structures can be regressive, which is why Denver’s Council refinements moved toward a flat residential rate with a low-income rebate.

Property tax instruments use millages or parcel taxes. Ann Arbor funds sidewalk repairs through a 0.125-mill property tax approved by voters in 2011. Berkeley passed Measure FF in November 2024, establishing a special parcel tax of \$0.17 per square foot on residential improvements and \$0.25 on other properties, generating an estimated \$15 million annually for fourteen years. Berkeley’s measure is technically a parcel tax rather than a millage, a distinction driven by California’s Proposition 13, which requires a two-thirds vote for property tax increases but only a simple majority for parcel taxes. State-level constraints of this kind shape the mechanisms available to municipalities, a point consistent with the home-rule discussion in Chapter 2. Tax-based models can generate substantial revenue but typically require voter approval, and millages also require periodic renewal.

Sidewalk Improvement Districts organize the city geographically and fund maintenance through annual fees scaled by property classification. Ithaca’s five SIDs charge a low base rate for single-family residential parcels and tiered fees for other property types, effectively socializing maintenance costs across all property owners within each district rather than placing the entire cost on the abutting parcel. Improvement districts can support equity scaling but require additional administrative infrastructure to operate.

Combined transportation funding approaches bundle sidewalks into a broader streets-and-sidewalks funding measure. East Grand Rapids uses a 2-mill levy covering both, Cheney funds streets and sidewalks through a 4 percent fee on electricity and natural gas bills, and Seattle’s transportation levy directs sidewalk investments using an equity index that prioritizes underserved neighborhoods (Seattle Department of Transportation, 2025). The combined approach produces a larger revenue pool, but sidewalks compete with street priorities within that pool. Seattle’s case carries an additional caveat: the levy funds new sidewalk construction, but the City still retains property-owner liability for maintenance. The levy reforms how new infrastructure is built rather than the underlying legal model governing the existing network. Each levy cycle, additionally, requires the city to return to voters for reauthorization, creating political uncertainty that complicates long-range planning.

The two-tier structure clarifies what reform actually requires. A city that adopts an inspection-and-bill program – or a shared-cost or point-of-sale program – is improving the operational delivery of the property-owner model without changing it. A city that adopts a utility fee, dedicated fee, millage, parcel tax, SID, or general-fund program is replacing the property-owner model with a different governance structure. Both are policy choices, and both are available to council members and practitioners considering reform. The distinction is whether the choice is operational or structural, and the evidence in this chapter and the next suggests that structural change is what produces a different outcome at the scale of an entire network.

Within either tier, several design decisions shape what a reform looks like in practice:

- **Revenue structure:** whether to fund sidewalks through the general budget, a dedicated fee, a tax measure, or a hybrid.
- **Authorization pathway:** whether reform can be achieved through council action (Payette, Cheney, Englewood) or requires a voter mandate (Denver, Ann Arbor, Berkeley).
 - Council-driven reforms are faster to implement but may face less public buy-in; voter-approved measures carry stronger political legitimacy but require sustained advocacy campaigns.
- **Cost distribution:** who pays, and whether the structure is progressive, regressive, or scaled to property type.
- **Physical infrastructure strategy:** whether the program is flexible enough to accommodate different levels of investment across neighborhood types, including interim mechanisms like Seattle’s walkway program for high-cost retrofit areas.
- **Scope of transfer:** whether the program addresses maintenance, new construction, or both. Lincoln’s partial transfer and Seattle’s levy-funded new construction illustrate that scope can be a deliberate design choice rather than an oversight.

Our guide presents each model as a route with distinct prerequisites, timelines, and tradeoffs rather than ranking them, allowing a reader to evaluate which pathway best fits their city’s political environment, legal authority, fiscal capacity, and stage of public awareness. The two-tier distinction is preserved in the guide so that practitioners and

council members can see clearly whether the option in front of them is an operational reform within the existing model or a transition to a new one. The dollar amounts and population figures associated with each example city are carried into the Story Map directly, so a reader can calibrate feasibility against cities at a comparable scale.

5.4 Equity as an Implementation Lens

The spatial findings in Section 4.3 confirm what national research has documented for over a decade: sidewalk infrastructure deficits are not evenly distributed, and they correlate with historical patterns of disinvestment. In Denver and Minneapolis, areas graded C and D under the HOLC redlining maps show significantly higher rates of missing sidewalks than areas graded A or B. This is consistent with the 2012 Bridging the Gap findings that low-income neighborhoods are less likely to have sidewalks and safety features (Bridging the Gap, 2012), and with the 2015 “Rethinking Denver Sidewalk Policy” brief, which reported that 47% of streets in low-income areas lacked adequate sidewalks and that low-income neighborhoods experienced more than double the pedestrian fatality rate of wealthier communities (WalkDenver & Mile High Connects, 2015a).

The data also reveals a finding that complicates a single-axis redlining narrative. As Section 4.3 documents, the largest single-category share of missing sidewalk in Denver and Seattle falls not in redlined neighborhoods themselves but in areas that HOLC never surveyed – 61% of Denver’s gaps and 62% of Seattle’s. Minneapolis shows a more even distribution, with 39% in ungraded areas and 38% in graded C and D areas combined. These ungraded areas typically correlate with land that was undeveloped or non-residential at the time of the surveys and later transitioned into car era suburban development. They inherited neither the pre-car era sidewalk networks of the urban core nor the scrutiny of the redlining maps. They represent a second, distinct equity layer: not the product of discriminatory lending but of planning philosophies that treated sidewalks as optional in lower-density, automobile-oriented development. Any equity-based prioritization framework that focuses exclusively on historically redlined areas will miss substantial concentrations of missing infrastructure, particularly in Denver and Seattle.

It is also worth noting that within Denver specifically, present-day equity priority areas – particularly on the city’s west side – extend well beyond what the standard “urban core” framing captures. A substantial share of Denver’s narrow or substandard sidewalks falls in neighborhoods that are equity priorities under the city’s current frameworks but that fall outside the older first-ring areas typically associated with the urban core. Citywide deferred maintenance compounds this pattern: rather than resources being directed away from older neighborhoods, sidewalk maintenance has been broadly underfunded across the city for decades, and the visible deficits today reflect that uniform underinvestment as much as any neighborhood-specific pattern.

The transit proximity analysis in Section 4.4 reinforces the multi-layer view. Even within a quarter mile of historic streetcar lines, the sidewalk network remained a patchwork that reflected the economic status of residents rather than the transportation needs of the corridor. Lower-income residents who depended on the streetcar often walked in the street or on unpaved paths to reach it because no municipal mandate required sidewalk

construction on their blocks. This historical pattern has direct implications for modern transit investment. As cities expand bus rapid transit, light rail, and other fixed-route services, the assumption that nearby sidewalk infrastructure already exists is often wrong, particularly in the neighborhoods most likely to depend on transit. Without coordinated investment in pedestrian access alongside transit expansion, cities risk replicating the same gap: building systems that serve some residents well while leaving others to navigate incomplete or nonexistent connections on foot.

Some peer cities have begun to incorporate equity into their implementation frameworks. Seattle's transportation levy directs sidewalk investments using an equity index that prioritizes underserved neighborhoods (Seattle Department of Transportation, 2025), and Minneapolis's Pedestrian Plan specifically targets sidewalk gap infill near schools and parks (City of Minneapolis, 2020). Denver's ArcGIS StoryMap, *Sidewalk Implementation Plan: State of the System*, shows that more than half of transit stops on high-ridership corridors are within a two-minute walk of a missing sidewalk, and more than a third of schools, parks, grocery stores, and health centers are in the same position (City and County of Denver, 2026). These proximity metrics offer a model for how cities can move beyond broad geographic equity categories and toward a needs-based prioritization that accounts for where people actually walk and what they are trying to reach.

The equity dimension also intersects with the funding models discussed in Section 5.3. Flat-fee structures, while administratively simple, place the same cost on low-income homeowners as on high-income ones. Denver recognized this in refining its post-Initiative 307 fee structure with a low-income rebate program, and Ithaca's tiered district model scales fees by property classification. Cities that pursue sidewalk reform without addressing the distributional impact of their funding mechanism risk creating a system that charges equitably but builds inequitably or vice versa. The guide uses this multi-layer equity framing throughout its prioritization discussion. Rather than treating equity as a standalone consideration, the guide presents it as a condition that shapes governance design, funding structure, and implementation sequencing simultaneously. The findings from this analysis suggest that an effective equity lens must account for at least three layers: the legacy of historical disinvestment captured in the HOLC grade analysis, the pedestrian infrastructure absence in car era neighborhoods that fall outside historically surveyed areas, and the ongoing gaps in pedestrian access near transit, schools, and essential services. Each layer is reflected in the guide's discussion of cost distribution, infrastructure flexibility, and equity-driven prioritization.

5.5 Conclusion

This report set out to document the background research, spatial analysis, and policy comparison behind the Sidewalk Network Reform Guide. The four preceding chapters of analysis converge on a single argument: the private-liability sidewalk model that prevails in most American cities is a policy choice, not a structural inevitability, and it is a choice that compounds its own costs over time. Each year a city defers the transition to coordinated sidewalk management, the infrastructure gap widens, the retrofit expense grows, the legal exposure under the ADA accumulates, and the equity consequences deepen. The path-dependency analysis in Section 5.1 explains why this

compounding occurs; the Denver reform trajectory in Section 5.2 illustrates how long it can take to reverse; the comparative typology in Section 5.3 distinguishes operational reform from structural reform and documents the mechanisms cities have used to make the structural transition; and the equity analysis in Section 5.4 establishes the cross-cutting condition under which any of those models must operate.

Denver's two-decade journey from first proposal to funded implementation plan is the most instructive contemporary case, but it is not the only template. Cities such as Payette, Englewood, and Cheney show that structural reform can begin with a single council vote; Berkeley and Ann Arbor demonstrate the path through a voter-approved tax instrument; Seattle's walkway program illustrates how interim infrastructure can deliver pedestrian access where full retrofit remains years away. Minneapolis, by contrast, illustrates the limit of operational reform within the property-owner model: a well-administered inspection-and-bill program improves the cadence of repairs but does not, on its own, close the network or shift the underlying liability. Taken together, these examples define a spectrum of reform along which any city can locate itself, identify the adjacent options, and select a pathway calibrated to its political, legal, and fiscal context.

Mind the Gap: Lessons from Denver to Rebuild Your Sidewalk Network translates the historical research, era-based spatial findings, governance typology, and equity framing developed across these chapters into an interactive website that practitioners can navigate without any expertise or prerequisite knowledge. The website carries the central finding outward: that sidewalk governance is a design problem with available solutions, and that the cost of inaction is concrete, measurable, and growing. The website is built to meet cities wherever they are in that process, whether they are just beginning to quantify their sidewalk gaps, building the political case for a funding mechanism, or preparing to put a measure before voters. [The website can be found here.](#)

Appendices

Appendix A: GIS Analysis Methodology

The spatial analysis for this study required a data normalization process to ensure parity between the sidewalk inventories of Denver, Seattle, and Minneapolis. Each municipality uses a different schema to track sidewalk infrastructure, so the first step was to reclassify the Denver and Seattle Sidewalk datasets. These inventories were simplified into fewer concise categories to maintain consistency with the Minneapolis Binary Model, which shows presence and absence of sidewalks. This standardization made a true comparison between sidewalk infrastructure possible.

To evaluate the influence of historical transit, the study used a proximity buffer centered on ¼ mile access to a historical streetcar line. By intersecting this zone with a sidewalk gap layer, we were able to quantify the percentage of missing infrastructure within historical transit corridors versus those outside the network's reach. For Minneapolis and Denver this involved using rail alignment datasets, while Seattle's layer was manually digitized and reconstructed based on a 1931 map of their streetcar system.

To determine the impact historical equity practices had on the sidewalk network, the project team downloaded redlining spatial data for each city found in the Mapping Inequality, University of Richmond webtool. These missing sidewalk segments were overlaid to find how significant the history of disinvestment has been on today's sidewalks. By isolating these historical boundaries, the analysis could measure the presence of sidewalk infrastructure deficits, providing a direct link between historical lending discrimination and current pedestrian inaccessibility.

The final spatial correlation relied on a year-built attribute join to categorize the sidewalk gaps into two development eras. While Denver and Seattle provided relatively integrated building age data, the process for Minneapolis required a more intensive data-linking workflow. Although Minneapolis provided parcel data, the information was not mapped. To resolve this, we utilized the Minneapolis city assessor's dataset to extract "Year Built" attributes and combined this with the city's land use map. Following this, parcels were divided at the 1940 threshold to distinguish between the pre-car era and car era, allowing for statistical comparison of how planning philosophies impacted the built environment.

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